

256Mb (16M x 16 bit) U ϵ RAM

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Document Title

16Mx16 bit Synchronous Burst Uni-Transistor Random Access Memory

Revision History

<u>Revision No.</u>	<u>History</u>	<u>Draft Date</u>	<u>Remark</u>
0.0	Initial	March 17, 2006	Preliminary

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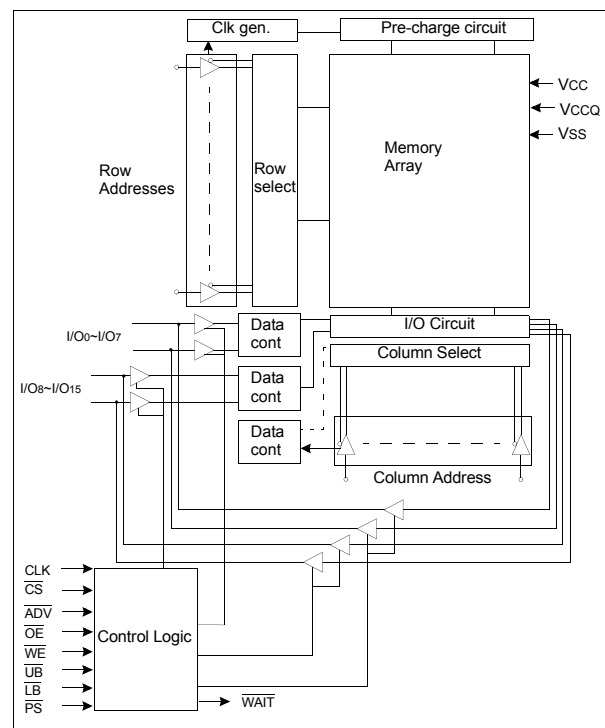
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16M x 16 bit Synchronous Burst Uni-Transistor CMOS RAM**GENERAL DESCRIPTION**

The world is moving into the mobile multi-media era and therefore the mobile handsets need bigger & faster memory capacity to handle the multi-media data. SAMSUNG's UtRAM products are designed to meet all the request from the various customers who want to cope with the fast growing mobile market. UtRAM is the perfect solution for the mobile market with its low cost, high density and high performance feature. K1B5616BAM is fabricated by SAMSUNG's advanced CMOS technology using one transistor memory cell. The device supports the traditional SRAM like asynchronous operation (asynchronous page read and asynchronous write), the NOR flash like synchronous operation (synchronous burst read and asynchronous write). These two operation modes are defined through the mode register setting. The device also supports the special features for the standby power saving. Those are the Partial Array Refresh(PAR) mode, Deep Power Down(DPD) mode and internal TCSR (Temperature Compensated Self Refresh). The optimization of output drive strength is possible through the mode register setting to adjust for the different data loadings. Through this drive strength optimization, the device can minimize the noise generated on the data bus during read operation.

FEATURES & FUNCTION BLOCK DIAGRAM

- Process technology: CMOS
- Organization: 16M x 16 bit
- Power supply voltage: 1.7V~1.95V
- Three state outputs
- Supports MRS (Mode Register Set)
 - $\overline{PS}$ pin set up
 - Software set up
- Supports power saving modes
 - PAR (Partial Array Refresh)
 - DPD (Deep Power Down)
 - Internal TCSR (Temperature Compensated Self Refresh)
- Supports driver strength optimization
- K1B5616BAM supports
 - Asynchronous read / Asynchronous write
 - Synchronous burst read / Asynchronous write
- Synchronous burst operation
 - Max. clock frequency : 104MHz
 - Fixed and Variable read latency
 - 4 / 8 / 16 / 32 and Continuous burst
 - Wrap / No-wrap
 - Latency : 4(Variable) @ 104MHz
 - 3(Variable) @ 80MHz
 - 2(Variable) @ 66MHz
- Burst stop
- Burst read suspend
- Burst write data masking

**PRODUCT FAMILY**

Product Family	Operating Mode ¹⁾	Operating Temp.	Vcc Range	Speed	Current Consumption	
					Standby (I _{SB1} , Max.)	Operating (I _{CC2P} , Max.)
K1B5616BAM-I	Mode 1 & Mode 2	Industrial(-25~85°C)	1.7~1.95V	104MHz	350μA < 85°C 200μA < 40°C	20mA

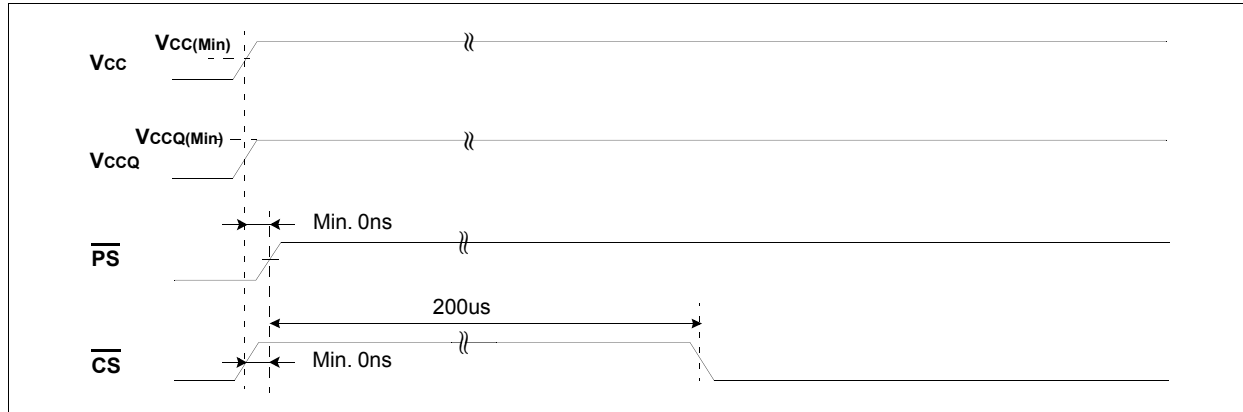
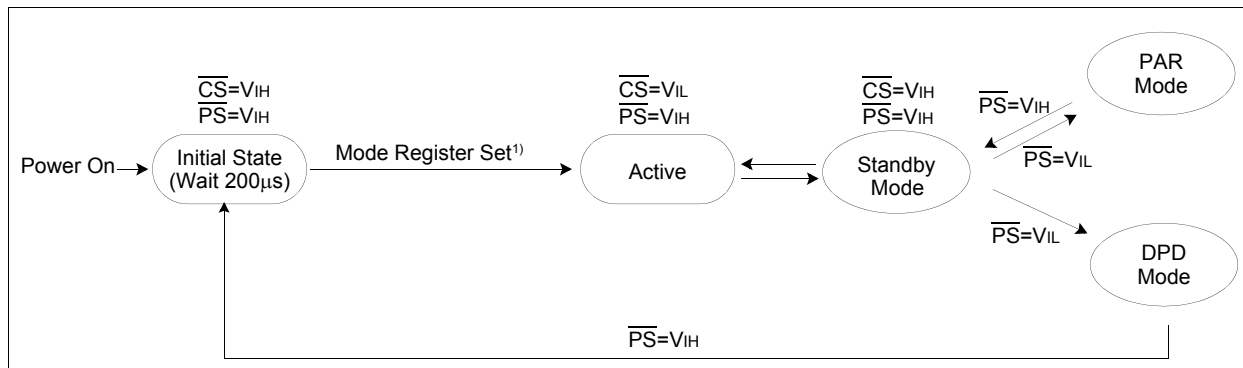
1) Mode1 : Asynchronous read / Asynchronous write
 Mode2 : Synchronous burst read / Asynchronous write

TERMINOLOGY DESCRIPTION

Name	Function	Type	Description
CLK	Clock	Input	Synchronizes the memory to the system operating frequency during synchronous operations. Commands are referenced to CLK.
$\overline{\text{ADV}}$	Address Valid	Input	Indicates that a valid address is present on the address inputs. Addresses can be latched on the rising edge of $\overline{\text{ADV}}$ during asynchronous READ and WRITE operations.
$\overline{\text{PS}}$	Mode Register set	Input	$\overline{\text{PS}}$ low enables Mode Register to be set and enables either PAR or DPD to be set.
$\overline{\text{CS}}$	Chip Select	Input	$\overline{\text{CS}}$ low enables the chip to be active $\overline{\text{CS}}$ high disables the chip and puts it into standby mode or deep power down mode.
$\overline{\text{OE}}$	Output Enable	Input	Enables the output buffers when LOW. when $\overline{\text{OE}}$ is HIGH, the output buffers are disabled.
$\overline{\text{WE}}$	Write Enable	Input	$\overline{\text{WE}}$ low enables the chip to start writing the data
$\overline{\text{LB}}$	Lower Byte (I/O _{0~7})	Input	$\overline{\text{UB}}$ ($\overline{\text{LB}}$) low enables upper byte (lower byte) to allow data Input/output from I/O buffers.
$\overline{\text{UB}}$	Upper Byte (I/O _{8~15})	Input	
A0~A23	Address 0 ~ Address 23	Input	Inputs for addresses during READ and WRITE operations. Addresses are internally latched during READ and WRITE cycles.
I/O _{0~15}	Data Inputs / Outputs	Input/Output	Depending on $\overline{\text{UB}}$ or $\overline{\text{LB}}$ status, word(16-bit, $\overline{\text{UB}}$ & $\overline{\text{LB}}$ low) data, upper byte(8-bit, $\overline{\text{UB}}$ low & $\overline{\text{LB}}$ high) data or lower byte(8-bit, $\overline{\text{LB}}$ low & $\overline{\text{UB}}$ high) data is loaded
VCC	Voltage Source	Power	Device Power supply. Power supply for device core operation.
VCCQ	I/O Voltage Source	Power	I/O Power supply. Power supply for input/output buffers.
VSS	Ground Source	GND	Ground for device core operation
VSSQ	I/O Ground Source	GND	Ground for input/output buffers
$\overline{\text{WAIT}}$	Valid Data Indicator	Output	The $\overline{\text{WAIT}}$ signal is output signal indicating the status of the data on the bus whether or not it is valid. $\overline{\text{WAIT}}$ is asserted when a burst crosses a word-line boundary. $\overline{\text{WAIT}}$ is asserted and should be ignored during asynchronous and page mode operations.

POWER UP SEQUENCE

After V_{CC} and V_{CCQ} reach minimum operating voltage(1.7V), drive $\overline{CS}$ High first and then drive $\overline{PS}$ High. Then the device gets into the Power Up mode. Wait for minimum 200 μ s to get into the normal operation mode. During the Power Up mode, the standby current can not be guaranteed. To get the stable standby current level, at least one cycle of active operation should be implemented regardless of wait time duration. To get the appropriate device operation, be sure to keep the following power up sequence. MODE1(Asynchronous Read / Asynchronous Write) is set up after power up, but this mode is not always guaranteed.

**MODE STATE MACHINE**

1) Refer to MRS(Mode Register Set).

ABSOLUTE MAXIMUM RATINGS

Item	Symbol	Ratings	Unit
Voltage on any pin relative to Vss	V _{IN} , V _{OUT}	-0.2 to V _{CCQ} +0.3V	V
Power supply voltage relative to Vss	V _{CC} , V _{CCQ}	-0.2 to 2.5V	V
Power Dissipation	P _D	1.0	W
Storage temperature	T _{STG}	-65 to 150	°C
Operating Temperature	T _A	-25 to 85	°C

1) Stresses greater than "Absolute Maximum Ratings" may cause permanent damage to the device. Functional operation should be restricted to be used under recommended operating condition. Exposure to absolute maximum rating conditions longer than 1 second may affect reliability.

RECOMMENDED DC OPERATING CONDITIONS

Item	Symbol	Min	Typ	Max	Unit
Power supply voltage(Core)	V _{CC}	1.7	1.8	1.95	V
Power supply voltage(I/O)	V _{CCQ}	1.7	1.8	1.95	V
Ground	V _{SS} , V _{SSQ}	0	0	0	V
Input high voltage	V _{IH}	0.8 x V _{CCQ}	-	V _{CCQ} +0.2 ²⁾	V
Input low voltage	V _{IL}	-0.2 ³⁾	-	0.4	V

1. T_A= -25 to 85°C, otherwise specified.

2. Overshoot: V_{CCQ} +1.0V in case of pulse width ≤20ns. Overshoot is sampled, not 100% tested.

3. Undershoot: -1.0V in case of pulse width ≤20ns. Undershoot is sampled, not 100% tested.

CAPACITANCE (f=1MHz, T_A=25°C)

Item	Symbol	Test Condition	Min	Max	Unit
Input capacitance	C _{IN}	V _{IN} =0V	-	8	pF
Input/Output capacitance	C _{IO}	V _{IO} =0V	-	8	pF

DC AND OPERATING CHARACTERISTICS

Item	Symbol	Test Conditions			Min	Typ	Max	Unit	
Input Leakage Current	I _{LI}	V _{IN} =V _{SS} to V _{CCQ}			-1	-	1	μA	
Output Leakage Current	I _{LO}	$\overline{CS}=V_{IH}$, $\overline{PS}=V_{IH}$, $\overline{OE}=V_{IH}$ or $\overline{WE}=V_{IL}$, V _{IO} =V _{SS} to V _{CCQ}			-1	-	1	μA	
Average Operating Current(Async)	I _{CC2} ⁶⁾	Cycle time=70ns, I _{IO} =0mA ⁴⁾ , 100% duty, $\overline{CS}=V_{IL}$, $\overline{PS}=V_{IH}$, V _{IN} =V _{IL} or V _{IH}			-	-	35	mA	
	I _{CC2P}	Cycle time=t _{RC} +3t _{PC} , I _{IO} =0mA ⁴⁾ , 100% duty, $\overline{CS}=V_{IL}$, $\overline{PS}=V_{IH}$, V _{IN} =V _{IL} or V _{IH}			-	-	20	mA	
Average Operating Current(Sync)	I _{CC3}	Burst Length 4, Latency 5, 80MHz, I _{IO} =0mA ⁴⁾ , Address transition 1 time, $\overline{CS}=V_{IL}$, $\overline{PS}=V_{IH}$, V _{IN} =V _{IL} or V _{IH}			-	-	35	mA	
Output Low Voltage	V _{OL}	I _{OL} =0.1mA			-	-	0.2	V	
Output High Voltage	V _{OH}	I _{OH} =-0.1mA			1.4	-	-	V	
Standby Current(CMOS)	I _{SB1} ¹⁾	$\overline{CS} \geq V_{CCQ}-0.2V$, $\overline{PS} \geq V_{CCQ}-0.2V$, Other inputs=V _{SS} or V _{CCQ} (Toggle is not allowed) ⁵⁾		< 40°C	-	-	200	μA	
				< 85°C	-	-	350	μA	
Partial Refresh Current	I _{SBP} ²⁾	$\overline{PS} \leq 0.2V$, $\overline{CS} \geq V_{CCQ}-0.2V$, Other inputs=V _{SS} or V _{CCQ} (Toggle is not allowed) ⁵⁾		< 40°C	1/2 Block	-	-	180	μA
					1/4 Block	-	-	150	
				< 85°C	1/2 Block	-	-	250	μA
					1/4 Block	-	-	220	
Deep Power Down Current	I _{SBD}	$\overline{PS} \leq 0.2V$, $\overline{CS} \geq V_{CCQ}-0.2V$, Other inputs=V _{SS} or		< 85°C		-	-	10	μA

1. I_{SB1} is measured 60ms after $\overline{CS}$ high. CLK should be fixed at high or at Low.

2. Full Array Partial Refresh Current(I_{SBP}) is same as Standby Current(I_{SB1}).

3. Internal TCSR (Temperature Compensated Self Refresh) is used to optimize refresh cycle below 40°C.

4. I_{IO}=0mA; This parameter is specified with the outputs disabled to avoid external loading effects.

5. V_{IN}=0V; all inputs should not be toggle.

6. Clock should not be inserted between ADV low and $\overline{WE}$ low during Write operation.

MRS (MODE REGISTER SET)

The mode registers store the values for the various modes to make UtRAM suitable for a various applications through MRS. There are two ways to perform MRS. One is PS pin MRS and the other is Software MRS. The mode registers have lots of fields and each field consists of several options. Refer to the Table below for detailed Mode Register Setting. A19~A23 addresses are "Don't care" in Mode Register Setting.

MRS CODE

MRS code consists of 12 categories and several options in each category. RARS, PARA, PAR and DPD are related to power saving, BL, WC, Latency, Wrap, WP, MS and IL are related to bus operation and DS is related to device output impedance.

Mode Register Setting according to field of function

Address	A18	A17~A16	A15~A14	A13	A12	A11~A9	A8	A7~A5	A4	A3	A2	A1~A0
Function	IL	DS	MS	WP	Wrap	Latency	WC	BL	DPD	PAR	PARA	PARS

Initial Latency		Driver Strength			Mode Select						
A18	IL	A17	A16	DS	A15	A14	MS				
0	Fixed	0	0	Full Drive	0	0	Mode 1(Async. 4 Page Read / Async. Write)				
1	Variable	0	1	1/2 Drive	0	1	Mode 2(Sync. Burst Read / Async. Write)				
		1	0	1/4 Drive							

WAIT Polarity		Wrap		Latency Count				Wait Configuration		Burst Length			
A13	WP ¹⁾	A12	Wrap	A11	A10	A9	Latency	A8	WC	A7	A6	A5	BL
0	Low Enable	0	Wrap	1	0	0	2	0	One clock prior	0	1	0	4 word
1	High Enable	1	No-Wrap	0	0	0	3	1	At data	0	1	1	8 word
				0	0	1	4			1	0	0	16 word
				0	1	0	5			1	0	1	32 word
				0	1	1	6			1	1	1	Continuous ²⁾
				1	0	1	7						
				1	1	0	8						
				1	1	1	9						

Deep Power Down		Partial Array Refresh		PAR Array		PAR Size		
A4	DPD	A3	PAR	A2	PARA	A1	A0	PARS
0	DPD Enable	0	PAR Enable	0	Bottom Array	0	0	Full Array
1	DPD Disable	1	PAR Disable	1	Top Array	1	0	1/2 Array
						1	1	1/4 Array

[Note]

- A19~A23 addresses are "Don't care" & reserved for future use.
- The modes are set automatically to default modes which are 4 Page Read and Asynchronous Write / DPD disable / PAR disable after power up or DPD exit.

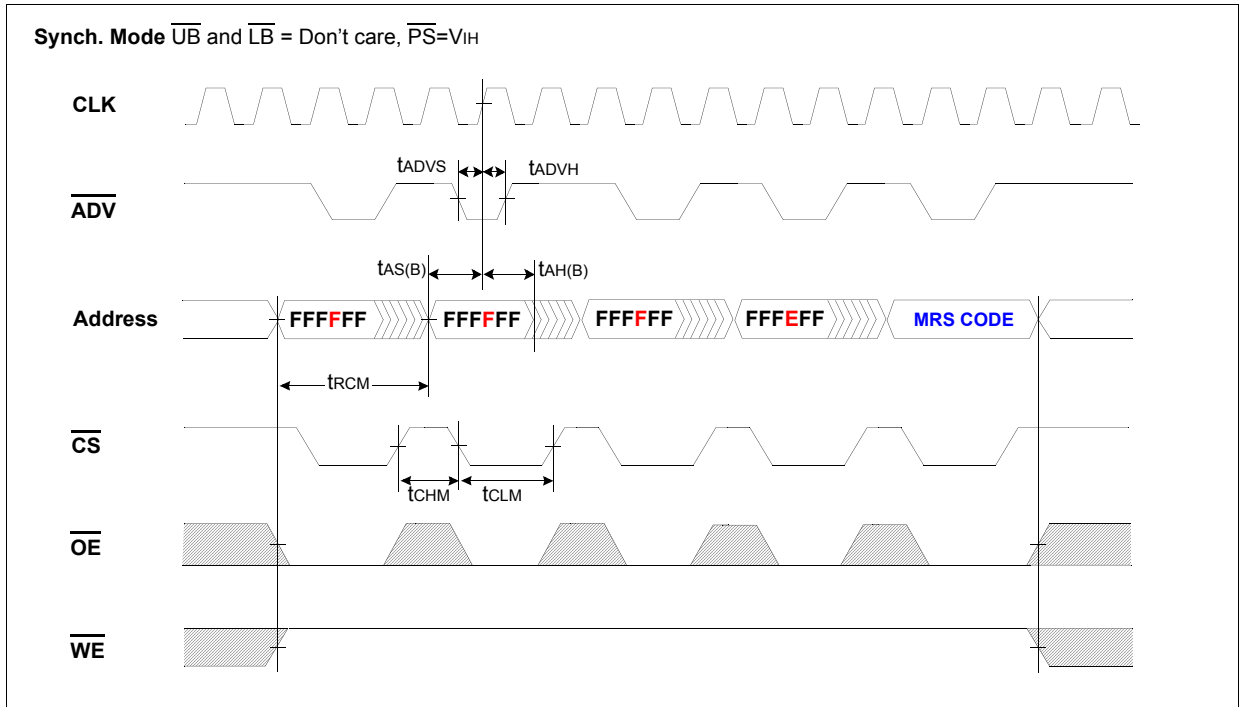
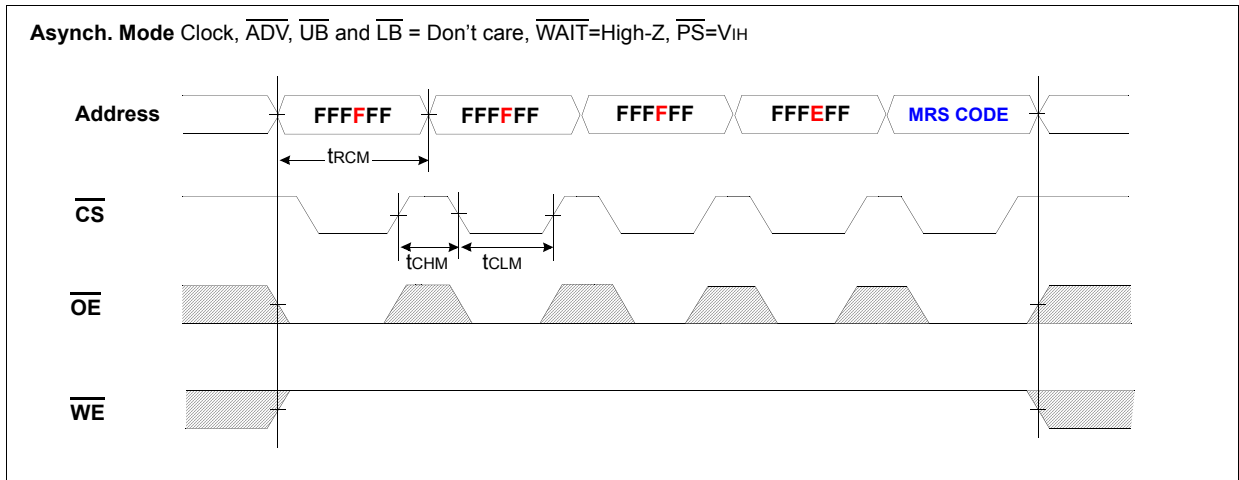
1) WP[0]; The data is available when $\overline{\text{WAIT}}$ signal is High. All the timings in this spec are illustrated based on this mode.

WP[1]; The data is available when $\overline{\text{WAIT}}$ signal is Low.

2) Refresh command will be denied during continuous operation. $\overline{\text{CS}}$ low should not be longer than tBC(max. 1.2us)

MRS TIMING WAVEFORM (SOFTWARE)

Software MRS timing consists of 5 Read cycles. Each cycle is normal Read cycle. $\overline{\text{CS}}$ pin should be toggling between cycles. 1st, 2nd and 3rd cycle should be FFFFFFFF(h), 4th cycle should be FFFEFF(h) and 5th cycle should be MRS code



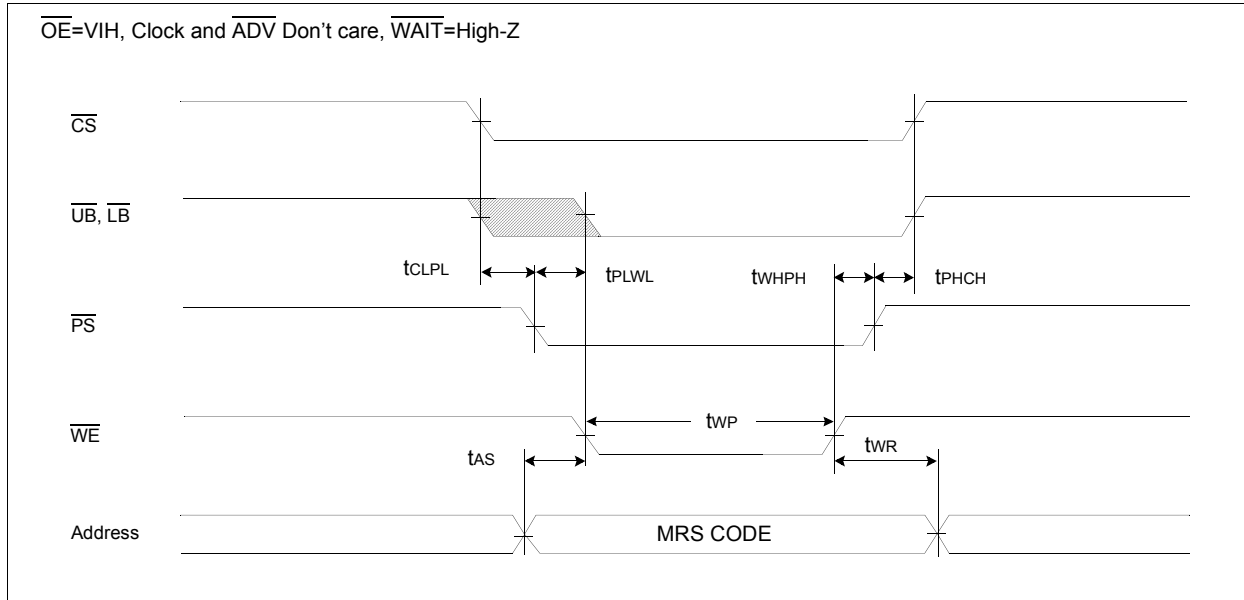
Note) Above timing and address condition should not be used in the normal operation. The above condition should be used only for the mode register setting purpose.

AC CHARACTERISTICS

Parameter List	Symbol	Min	Max	Units	Parameter List	Symbol	Min	Max	Units
$\overline{\text{ADV}}$ setup time to clock	t_{ADVS}	3	-	ns	Read cycle time	t_{RCM}	70	-	ns
$\overline{\text{ADV}}$ hold time from clock	t_{ADVH}	2	-	ns	$\overline{\text{CS}}$ high time	t_{CHM}	10	-	ns
Address setup time to clock	$t_{\text{AS(B)}}$	3	-	ns	$\overline{\text{CS}}$ low time	t_{CLM}	60	-	ns
Address hold time from clock	$t_{\text{AH(B)}}$	2	-	ns					

MRS TIMING WAVEFORM ($\overline{\text{PS}}$ Pin)

MRS can be implemented using by $\overline{\text{PS}}$ pin. Serial assertion of control signals of $\overline{\text{CS}}$, $\overline{\text{UB}}$ & $\overline{\text{LB}}$, $\overline{\text{PS}}$ and $\overline{\text{WE}}$ will get the device to be ready for MRS. MRS CODE should be set up before $\overline{\text{WE}}$ low and keep the CODE until one of those control signals deserts. MRS terminates when one of those control signals deserts. Clock & $\overline{\text{ADV}}$ are don't care in Asynchronous mode.

**AC CHARACTERISTICS**

Parameter List		Symbol	Speed		Units
			Min	Max	
MRS	$\overline{\text{CS}}$ Low to $\overline{\text{PS}}$ Low	tCLPL	0	-	ns
	$\overline{\text{PS}}$ Low to $\overline{\text{WE}}$ Low	tPLWL	0	-	ns
	$\overline{\text{WE}}$ High to $\overline{\text{PS}}$ High	tWHPH	0	-	ns
	$\overline{\text{PS}}$ High to $\overline{\text{CS}}$ High	tPHCH	0	-	ns

PAR (Partial Array Refresh) mode [A3~A1]

User can select half array, a fourth array as active memory array. The active memory array is periodically refreshed(data stored), whereas the disabled array is not going to be refreshed and so the previously stored data will be invalid. When re-enabling additional portions of the array, the new portions are available immediately upon writing to the MRS.

PAR mode execution;

- 1) Mode Register Setting into PAR enable(A3=0)

DPD enabled setting(A4=0) has higher priority to PAR enabled setting(A3=0). A4=1 is necessary to use PAR mode.

- 2) PAR mode Enter; keep $\overline{PS}$ signal at V_{IL} for longer than $0.5\mu s$ during standby mode (Mode Register: A4=1 & A3=0).
- 3) PAR mode Exit; The device returns to the standby mode when $\overline{PS}$ signal goes to V_{IH} during PAR mode.

* Mode register values are not changed after the device has been to PAR mode.

DPD (Deep Power Down) mode [A4]

The deep power down mode disables all the refresh related activities. This mode can be used when the system needs to save power. The data become invalid when DPD mode is executed.

DPD mode execution ;

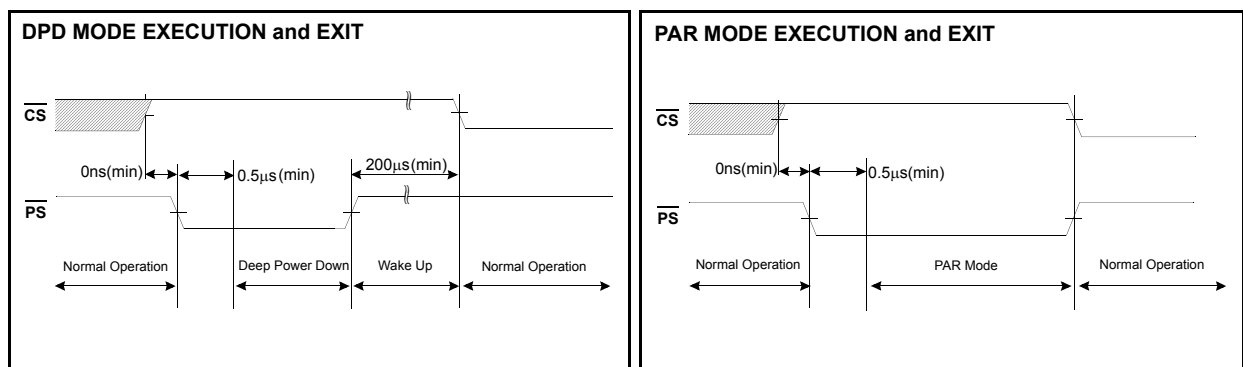
- 1) Mode Register Setting into DPD enable(A4=0)

- 2) DPD mode Enter; keep $\overline{PS}$ signal at V_{IL} for more than $0.5\mu s$ during standby mode (Mode Register: A4=0).

- 3) DPD mode Exit; The device returns to initial State when $\overline{PS}$ signal goes to V_{IH} during DPD mode. Wake up sequence is needed for the device to do normal operation.

* Mode register values are initialized to default value after the device has been to DPD mode.

Default modes are 4 Page Read and Asynchronous Write / DPD disable / PAR disable.

**STANDBY MODE CHARACTERISTICS**

Power Mode	Address (Bottom Array) ²⁾	Address (Top Array) ²⁾	Memory Cell Data	Standby ³⁾ (I_{SB1} , $<40^{\circ}C$)	Standby ³⁾ (I_{SB1} , $<85^{\circ}C$)	Wait Time(μs)
Standby(Full Array)	000000h ~ FFFFFFFh	000000h ~ FFFFFFFh	Valid ¹⁾	TBD	TBD	0
Partial Refresh(1/2 Block)	000000h ~ 7FFFFFFh	800000h ~ FFFFFFFh	Valid ¹⁾	TBD	TBD	0
Partial Refresh(1/4 Block)	000000h ~ 3FFFFFFh	C00000h ~ FFFFFFFh	Valid ¹⁾	TBD	TBD	0
Deep Power Down	000000h ~ FFFFFFFh		Invalid	TBD	TBD	200

1. Only the data in the selected block are valid

2. PAR Array can be selected through Mode Register Set

3. Standby mode is supposed to be set up after at least one active operation after power up.

I_{SB1} is measured after 60ms from the time when standby mode is set up.

Burst Length [A7~A5] & Wrap [A12]

The device supports 4 word, 8 word, 16 word, 32 word and Continuous burst read or write. and Wrap & No-Wrap are supported for Burst sequence.

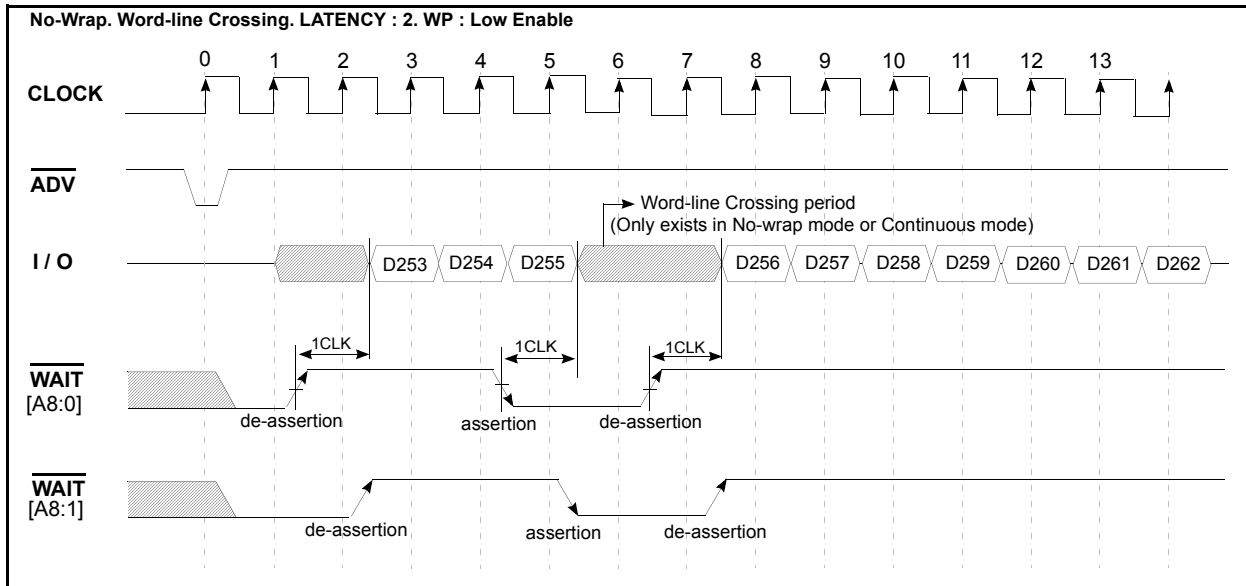
Burst Address Sequence(Decimal)					
Mode	Start	4 word	8 word	16 word	32 word
WRAP	0	0-1-2-3	0-1-2-3-4-5-6-7	0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15	0 - 1 - 2 - 3 - 4 - 5 ~ 26-27-28-29-30-31
	1	1-2-3-0	1-2-3-4-5-6-7-0	1-2-3-4-5-6-7-8-9-10-11-12-13-14-15-0	1 - 2 - 3 - 4 - 5 - 6 ~ 27-28-29-30-31 - 0
	2	2-3-0-1	2-3-4-5-6-7-0-1	2-3-4-5-6-7-8-9-10-11-12-13-14-15-0-1	2 - 3 - 4 - 5 - 6 - 7 ~ 28-29-30-31 - 0 - 1
	3	3-0-1-2	3-4-5-6-7-0-1-2	3-4-5-6-7-8-9-10-11-12-13-14-15-0-1-2	3 - 4 - 5 - 6 - 7 - 8 ~ 29-30-31-0-1-2
	~		~	~	~
	7		7-0-1-2-3-4-5-6	7-8-9-10-11-12-13-14-15-0-1-2-3-4-5-6	7 - 8 - 9 - 10 - 11 - 12 ~ 2 - 3 - 4 - 5 - 6
	~			~	~
	15			15-0-1-2-3-4-5-6-7-8-9-10-11-12-13-14	15-16-17-18-19-20 ~ 10-11-12-13-14
No-WRAP	~				~
	31				31-0-1-2-3-4 ~ 25-26-27-28-29-30
	0	0-1-2-3	0-1-2-3-4-5-6-7	0-1-2-3-4-5-6-7-8-9-10-11-12-13-14-15	0 - 1 - 2 - 3 - 4 - 5 ~ 26-27-28-29-30-31
	1	1-2-3-4	1-2-3-4-5-6-7-8	1-2-3-4-5-6-7-8-9-10-11-12-13-14-15-16	1 - 2 - 3 - 4 - 5 - 6 ~ 27-28-29-30-31-32
	2	2-3-4-5	2-3-4-5-6-7-8-9	2-3-4-5-6-7-8-9-10-11-12-13-14-15-16-17	2 - 3 - 4 - 5 - 6 - 7 ~ 28-29-30-31-32-33
	3	3-4-5-6	3-4-5-6-7-8-9-10	3-4-5-6-7-8-9-10-11-12-13-14-15-16-17-18	3 - 4 - 5 - 6 - 7 - 8 ~ 29-30-31-32-33-34
	~		~	~	~
	7		7-8-9-10-11-12-13-14	7-8-9-10-11-12-13-14-15-16-17-18-19-20-21-22	7 - 8 - 9 - 10-11-12 ~ 33-34-35-36-37-38
No-WRAP	~			~	~
	15			15-16-17-18-19-20-21-22-23-24-25-26-27-28-29-30	15-16-17-18-19-20 ~ 41-42-43-44-45-46
	~				~
	~				~
	31				31-32-33-34-35-36 ~ 57-58-59-60-61-62

1. Continuous Burst mode needs to meet tBC(max. 1.2us) parameter.

WAIT Configuration [A8] & WAIT Polarity [A13]

The WAIT signal is output signal indicating the status of the data on the bus whether or not it is valid. WAIT configuration is to decide the timing when WAIT asserts or deasserts. WAIT asserts (or deasserts) one clock prior to the data when A8 is set to 0. (WAIT asserts (or deasserts) at data clock when A8 is set to 1). WAIT polarity is to decide the WAIT signal level at which data is valid or invalid. Data is valid if WAIT signal is high when A13 is set to 0. (Data is valid if WAIT signal is low when A13 is set to 1). All the timing diagrams in this SPEC are illustrated based on following setup; [A13 : 0] and [A8 : 0].

Below timing shows WAIT signal's movement when word boundary crossing happens in No-wrap mode.

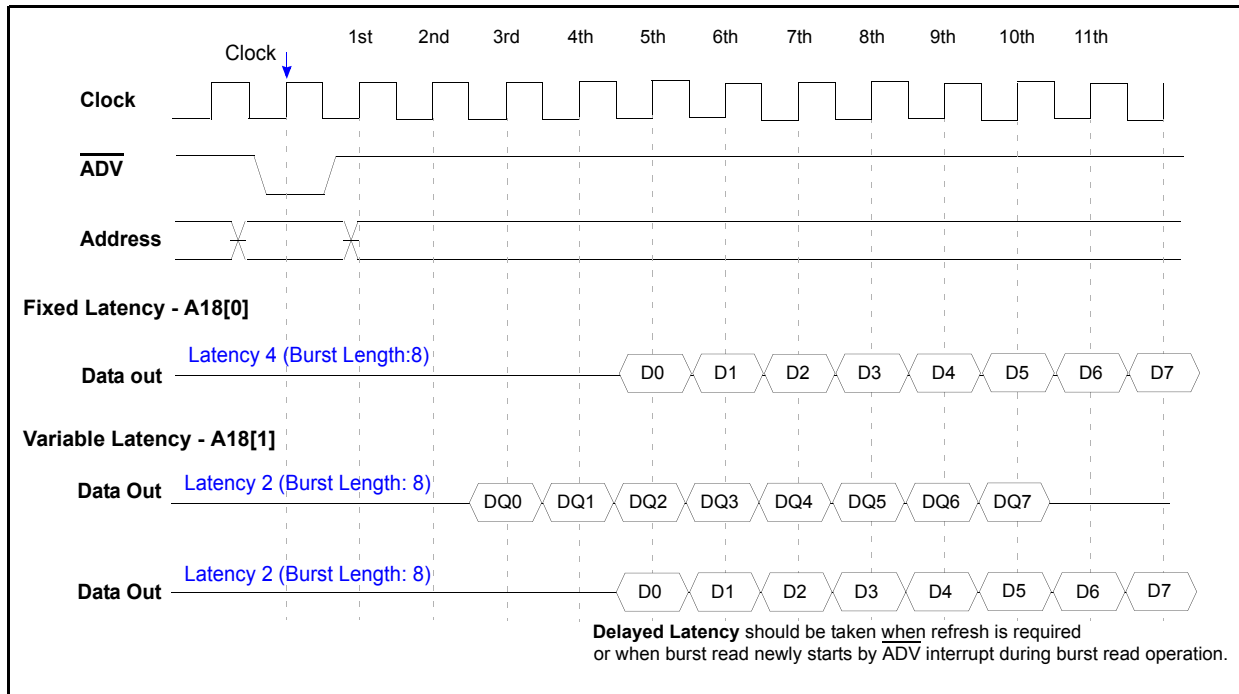


Latency [A11~A9]

The Latency stands for the number of clocks before the first data available from the burst command.

Item	Upto 66MHz		Upto 80MHz		Upto 104MHz	
	Fixed	Variable	Fixed	Variable	Fixed	Variable
Latency Set(A11:A10:A9)	4(0:0:1)	2(1:0:0)	5(0:1:0)	3(0:0:0)	7(1:0:1)	4(0:0:1)
Read Latency(min)	4	2 / 4 ¹⁾	5	3 / 5 ¹⁾	7	4 / 7 ¹⁾
1st Read data fetch clock	5th	3rd / 5th ¹⁾	6th	4th / 6th ¹⁾	8th	5th / 8th ¹⁾

1) Delayed Latency should be taken when refresh is required or when burst read newly starts by ADV interrupt during burst read operation.

**Driver Strength [A17~A16]**

The optimization of output driver strength is possible to adjust for the different data loadings. The device can minimize the noise generated on the data bus during read operation. The device supports full, 1/2 and 1/4 driver strength. The device's default mode is Full driver strength.

Driver Strength	Full	1 / 2	1 / 4
IMPEDANCE(typ.)	40Ω	90Ω	150Ω

1. Impedance values are typical values, not 100% tested.

OPERATION MODE [A15~A14]

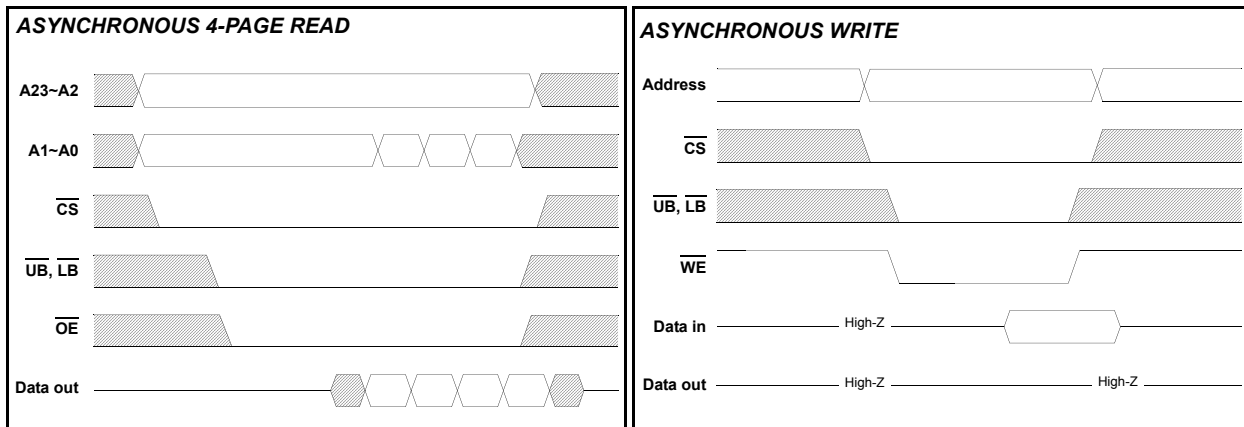
MODE1. ASYNCHRONOUS READ / ASYNCHRONOUS WRITE MODE

Asynchronous read operation

Asynchronous read operation starts when $\overline{CS}$, $\overline{OE}$ and $\overline{UB}$ or $\overline{LB}$ are asserted. First data come out after random access time(t_{AA}) but second, third and fourth data come out after page access time(t_{PA}) when using the page addresses (A0, A1). $\overline{PS}$ and $\overline{WE}$ should be de-asserted during read operation. Clock, $\overline{ADV}$ are don't care during read operation and $\overline{WAIT}$ is Hi-Z.

Asynchronous write operation

Asynchronous write operation starts when $\overline{CS}$, $\overline{WE}$ and $\overline{UB}$ or $\overline{LB}$ are asserted. $\overline{PS}$ should be de-asserted during write operation. Clock, $\overline{OE}$, $\overline{ADV}$ are don't care during write operation and $\overline{WAIT}$ signal is Hi-Z.



FUNCTIONAL DESCRIPTION

$\overline{CS}$	$\overline{PS}$	$\overline{OE}$	$\overline{WE}$	$\overline{LB}$	$\overline{UB}$	I/O0~7	I/O8~15	Mode	Power
H	H	X ¹⁾	X ¹⁾	X ¹⁾	X ¹⁾	High-Z	High-Z	Deselected	Standby
H	L	X ¹⁾	X ¹⁾	X ¹⁾	X ¹⁾	High-Z	High-Z	Deselected	DPD or PAR
L	H	H	H	X ¹⁾	X ¹⁾	High-Z	High-Z	Output Disabled	Active
L	H	X ¹⁾	X ¹⁾	H	H	High-Z	High-Z	Output Disabled	Active
L	H	L	H	L	H	Dout	High-Z	Lower Byte Read	Active
L	H	L	H	H	L	High-Z	Dout	Upper Byte Read	Active
L	H	L	H	L	L	Dout	Dout	Word Read	Active
L	H	H	L	L	H	Din	High-Z	Lower Byte Write	Active
L	H	H	L	H	L	High-Z	Din	Upper Byte Write	Active
L	H	H	L	L	L	Din	Din	Word Write	Active

1. X means "Don't care". X should be low or high state.

2. In asynchronous mode, Clock and $\overline{ADV}$ are ignored. Clock and $\overline{ADV}$ should be low or high state.

3. $\overline{WAIT}$ pin is High-Z in Asynchronous mode.

4. This mode(Mode 1) is supported K1B5616BAM product.

MODE2. SYNCHRONOUS BURST READ / ASYNCHRONOUS WRITE MODE

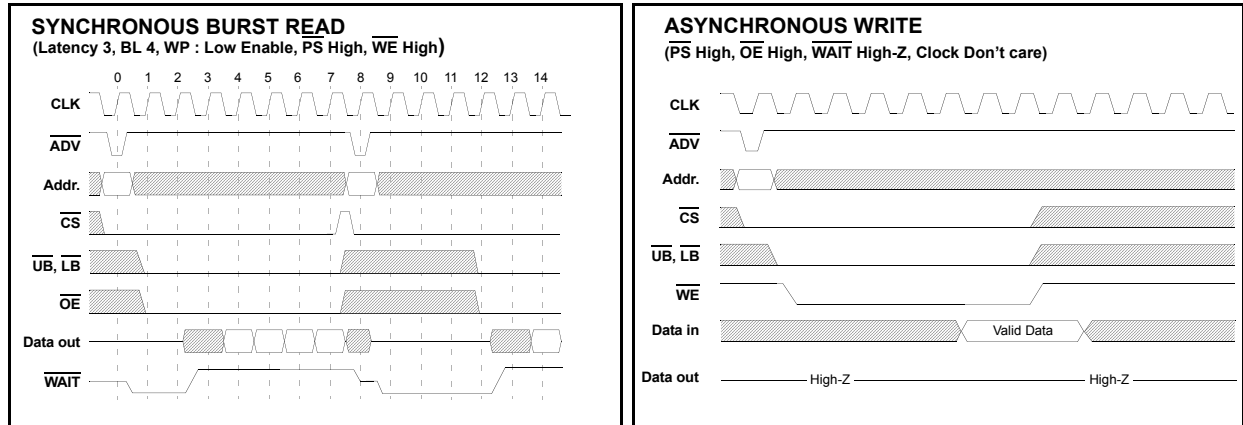
Synchronous Burst Read Operation

Burst Read command is implemented when $\overline{ADV}$ is detected low at clock rising edge. $\overline{WE}$ should be de-asserted during Burst read, Burst operation re-starts whenever $\overline{ADV}$ is detected low at clock rising edge even in the middle of operation.

Variable latency allows the UtRAM to be configured for minimum latency at high frequencies, but the controller must monitor $\overline{WAIT}$ to detect any conflict with refresh cycles.

Asynchronous Write Operation

Asynchronous write operation starts when $\overline{CS}$, $\overline{WE}$ and $\overline{UB}$ or $\overline{LB}$ are asserted. $\overline{PS}$ and should be de-asserted during write operation. Clock, $\overline{OE}$, $\overline{ADV}$ are don't care during write operation and $\overline{WAIT}$ signal is Hi-Z.



FUNCTIONAL DESCRIPTION

$\overline{CS}$	$\overline{PS}$	$\overline{OE}$	$\overline{WE}$	$\overline{LB}$	$\overline{UB}$	I/O0~7	I/O8~15	CLK	$\overline{ADV}$	Mode	Power
H	H	X ⁽¹⁾	X ⁽¹⁾	X ⁽¹⁾	X ⁽¹⁾	High-Z	High-Z	X ⁽²⁾	X ⁽²⁾	Deselected	Standby
H	L	X ⁽¹⁾	X ⁽¹⁾	X ⁽¹⁾	X ⁽¹⁾	High-Z	High-Z	X ⁽²⁾	X ⁽²⁾	Deselected	PAR
L	H	H	H	X ⁽¹⁾	X ⁽¹⁾	High-Z	High-Z	X ⁽²⁾	H	Output Disabled	Active
L	H	X ⁽¹⁾	X ⁽¹⁾	H	H	High-Z	High-Z	X ⁽²⁾	H	Output Disabled	Active
L	H	X ⁽¹⁾	H	X ⁽¹⁾	X ⁽¹⁾	High-Z	High-Z			Read Command	Active
L	H	L	H	L	H	Dout	High-Z		H	Lower Byte Read	Active
L	H	L	H	H	L	High-Z	Dout		H	Upper Byte Read	Active
L	H	L	H	L	L	Dout	Dout		H	Word Read	Active
L	H	H	L	L	H	Din	High-Z	X ⁽²⁾	or	Lower Byte Write	Active
L	H	H	L	H	L	High-Z	Din	X ⁽²⁾	or	Upper Byte Write	Active
L	H	H	L	L	L	Din	Din	X ⁽²⁾	or	Word Write	Active

1. X means "Don't care". X should be low or high state.

2. $\overline{WAIT}$ is device output signal so does not have any affect to the mode definition. Please refer to each timing diagram for $\overline{WAIT}$ pin function.

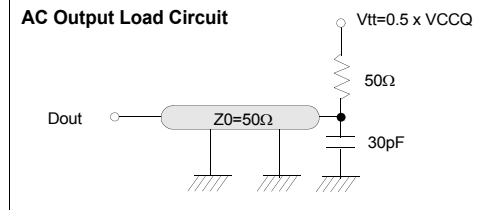
3. This mode(Mode 2) is supported only at K1B5616BAM product.

MODE 1 AC OPERATING CONDITIONS (ASYNCH. READ / ASYNCH. WRITE)**TEST CONDITIONS**

(Test Load and Test Input/Output Reference)

Input pulse level: 0.2 to V_{CCQ} -0.2V

Input rising and falling time: 3ns

Input and output reference voltage: 0.5 x V_{CCQ} Output load: $C_L=30\text{pF}$ V_{CC} : 1.7V~1.95V T_A : -25°C~85°C**AC Output Load Circuit****AC CHARACTERISTICS**

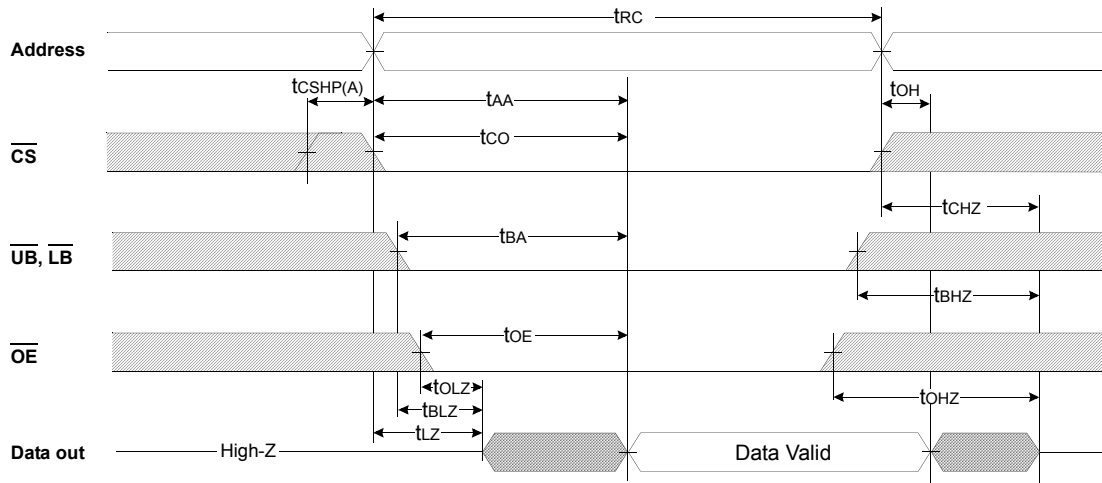
Parameter List		Symbol	Speed		Units
			Min	Max	
Common	$\overline{CS}$ High Pulse Width	$t_{CSHP(A)}$	10	-	ns
Asynch. Read	Read Cycle Time	t_{RC}	70	-	ns
	Page Read Cycle Time	t_{PC}	20	-	ns
	Address Access Time	t_{AA}	-	70	ns
	Page Access Time	t_{PA}	-	20	ns
	Chip Select to Output	t_{CO}	-	70	ns
	Output Enable to Valid Output	t_{OE}	-	20	ns
	$\overline{UB}$, $\overline{LB}$ Access Time	t_{BA}	-	20	ns
	Chip Select to Low-Z Output	t_{LZ}	10	-	ns
	$\overline{UB}$, $\overline{LB}$ Enable to Low-Z Output	t_{BLZ}	5	-	ns
	Output Enable to Low-Z Output	t_{OLZ}	5	-	ns
	Chip Disable to High-Z Output	t_{CHZ}	0	10	ns
	$\overline{UB}$, $\overline{LB}$ Disable to High-Z Output	t_{BHZ}	0	10	ns
	Output Disable to High-Z Output	t_{OHZ}	0	10	ns
	Output Hold	t_{OH}	5	-	ns
Asynch. Write	Write Cycle Time	t_{WC}	70	-	ns
	Chip Select to End of Write	t_{CW}	60	-	ns
	Address Set-up Time to Beginning of Write	t_{AS}	0	-	ns
	Address Valid to End of Write	t_{AW}	60	-	ns
	$\overline{UB}$, $\overline{LB}$ Valid to End of Write	t_{BW}	60	-	ns
	Write Pulse Width	t_{WP}	55 ¹⁾	-	ns
	$\overline{WE}$ High Pulse Width	t_{WHP}	5	-	ns
	Write Recovery Time	t_{WR}	0	-	ns
	Data to Write Time Overlap	t_{DW}	20	-	ns
	Data Hold from Write Time	t_{DH}	0	-	ns

1. $t_{WP}(\text{min})=70\text{ns}$ for continuous write without CS toggling longer than 1.2 μs 2. The High-Z timings measure a 100mV transition from either V_{OH} or V_{OL} toward $V_{CCQ} \times 0.5$ 3. The Low-Z timings measure a 100mV transition away from the High-Z level toward either V_{OH} or V_{OL} .

TIMING WAVEFORMS (ASYNCH. READ / ASYNCH. WRITE)

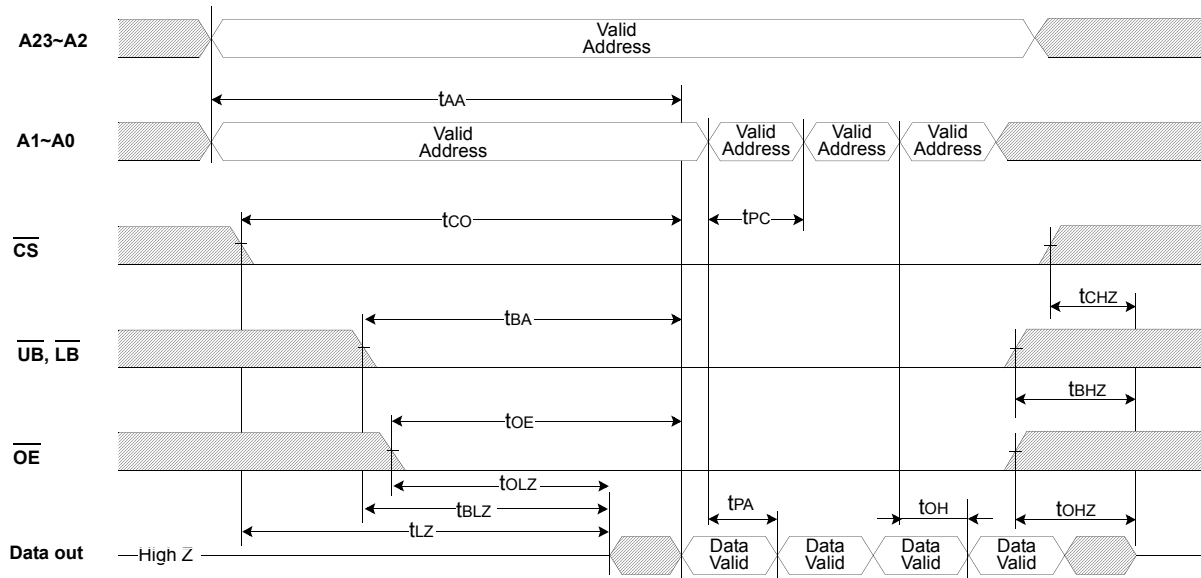
Asynch. READ

(PS=VIH, WE=VIH, WAIT=High-Z)



Asynch. PAGE READ

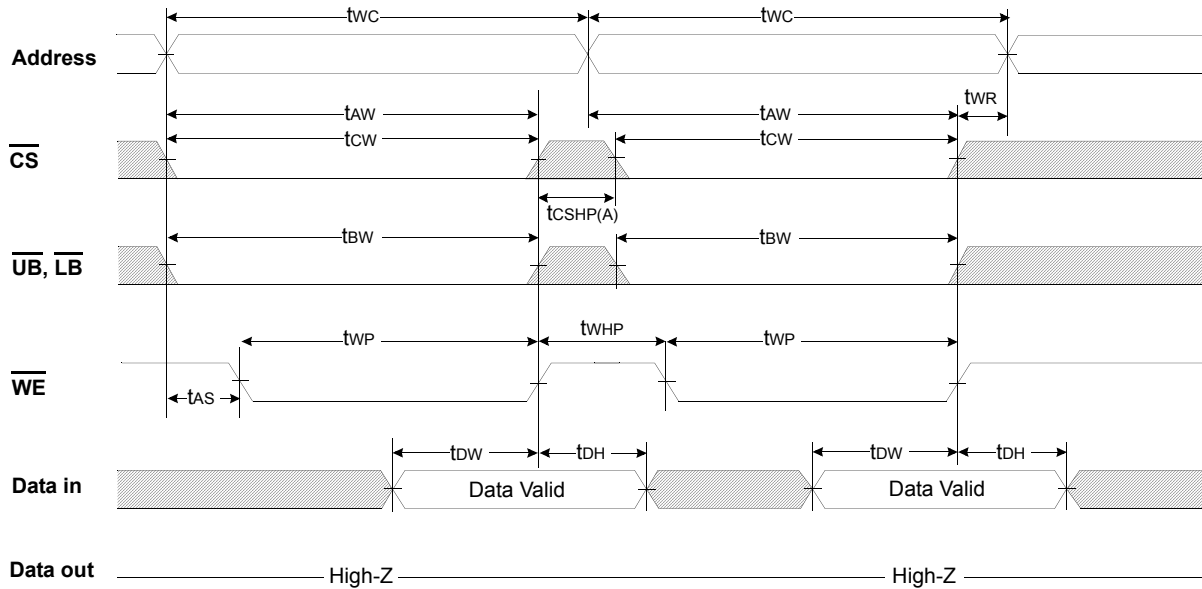
(PS=VIH, WE=VIH, WAIT=High-Z)



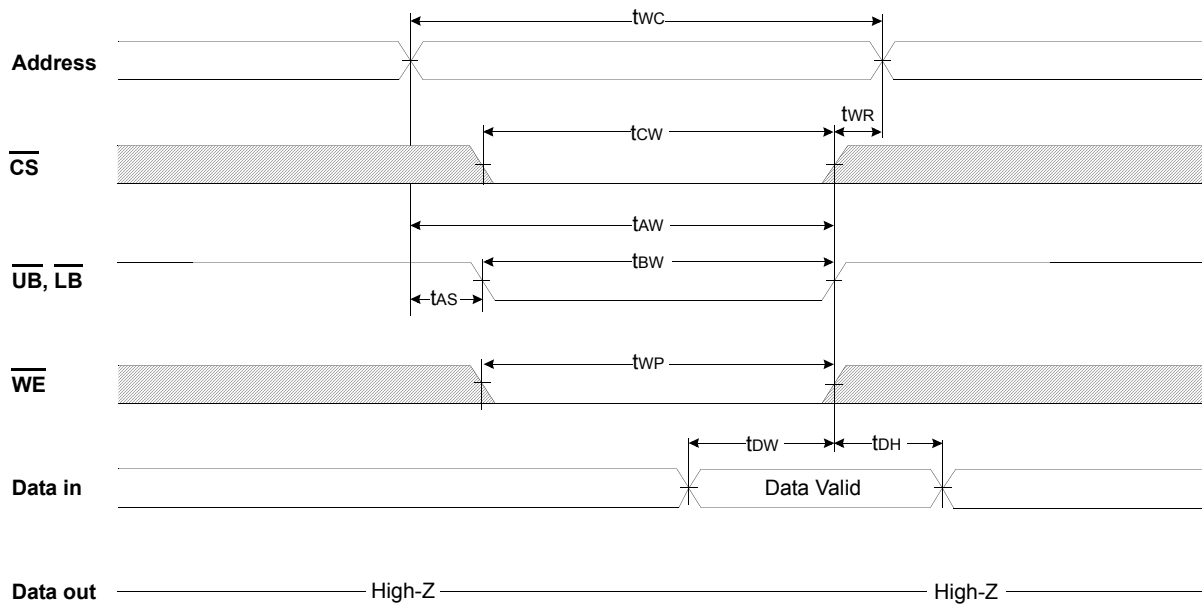
1. t_{CHZ} and t_{OHZ} are defined as the time at which the outputs achieve the open circuit conditions and are not referenced to output voltage levels.
2. At any given temperature and voltage condition, $t_{CHZ}(\text{Max.})$ is less than $t_{LZ}(\text{Min.})$ both for a given device and from device to device interconnection.
3. In asynchronous read cycle, Clock and $\overline{ADV}$ signals are ignored.
4. If invalid address signals shorter than min. t_{RC} are continuously repeated for over 1.2 μ s, the device needs a normal read timing(t_{RC}) or needs to sustain standby state for min. t_{RC} at least once in every 1.2 μ s.
5. In asynchronous 4 page read cycle, Clock and $\overline{ADV}$ signals are ignored.

Asynch. WRITE (1)

(PS=VIH, OE=VIH, WAIT=High-Z, WE Controlled)

**Asynch. WRITE (2)**

(PS=VIH, OE=VIH, WAIT=High-Z, UB & LB Controlled)



1. A write occurs during the overlap(t_{WP}) of low $\overline{CS}$ and low $\overline{WE}$. A write begins when $\overline{CS}$ goes low and $\overline{WE}$ goes low with asserting $\overline{UB}$ or $\overline{LB}$ for single byte operation or simultaneously asserting $\overline{UB}$ and $\overline{LB}$ for double byte operation. A write ends at the earliest transition when $\overline{CS}$ goes high or $\overline{WE}$ goes high. The t_{WP} is measured from the beginning of write to the end of write.
2. t_{CW} is measured from the $\overline{CS}$ going low to the end of write.
3. t_{AS} is measured from the address valid to the beginning of write.
4. t_{WR} is measured from the end of write to the address change. t_{WR} is applied in case a write ends with $\overline{CS}$ or $\overline{WE}$ going high.
5. In asynchronous write cycle, Clock and $\overline{ADV}$ signals are ignored.
6. Condition for continuous write operation over 15 times : $t_{WP}(\min)=70ns$

MODE 2 AC OPERATING CONDITIONS (SYNCH. READ / ASYNCH. WRITE)

TEST CONDITIONS

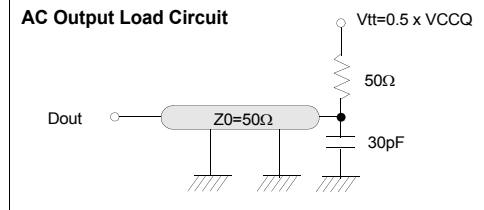
(Test Load and Test Input/Output Reference)

Input pulse level: 0.2 to V_{CCQ} -0.2V

Input rising and falling time: 1ns

Input and output reference voltage: 0.5 x V_{CCQ} Output load: $C_L=30\text{pF}$ V_{CC} : 1.7V~1.95V T_A : -25°C~85°C

AC Output Load Circuit



AC CHARACTERISTICS

Parameter List		Symbol	66MHz		80MHz		104MHz		Units
			Min	Max	Min	Max	Min	Max	
Synch. Burst Read	Clock Cycle Time	T	15	200	12.5	200	9.6	200	ns
	Burst Cycle Time	t _{BC}	-	1200	-	1200	-	1200	ns
	Address Set-up Time to clock	t _{AS(B)}	3	-	3	-	3	-	ns
	Address Hold Time from clock	t _{AH(B)}	2	-	2	-	2	-	ns
	ADV Setup Time to clock	t _{ADVS}	3	-	3	-	3	-	ns
	ADV Hold Time from clock	t _{ADVH}	2	-	2	-	2	-	ns
	CS Setup Time to clock	t _{CSS(B)}	3	-	3	-	3	-	ns
	CS High to ADV Low (Burst Stop)	t _{BSADV} ¹⁾	0	-	0	-	0	-	ns
	CS Low Hold Time from Clock(Burst Stop)	t _{CSLH}	2	-	2	-	2	-	ns
	CS High Pulse Width	t _{CSHP}	5	-	5	-	5	-	ns
	CS Low to WAIT Low	t _{WL}	-	12	-	12	-	12	ns
	Clock to WAIT High	t _{WH}	-	11	-	9	-	7	ns
	CS High to WAIT High-Z	t _{WZ}	-	10	-	10	-	10	ns
	UB, LB Low to End of Latency Clock	t _{BEL}	20	-	20	-	20	-	ns
	OE Low to End of Latency Clock	t _{OEL}	20	-	20	-	20	-	ns
	UB, LB Low to Low-Z Output	t _{BLZ}	5	-	5	-	5	-	ns
	OE Low to Low-Z Output	t _{OLZ}	5	-	5	-	5	-	ns
	Clock Rising to Data Output	t _{CD}	-	11	-	9	-	7	ns
	Output Hold from clock	t _{OH(B)}	2	-	2	-	2	-	ns
	Burst End Clock to Output High-Z	t _{HZ}	-	10	-	10	-	10	ns
	CS High to Output High-Z	t _{CHZ}	-	10	-	10	-	10	ns
	OE High to Output High-Z	t _{OHZ}	-	10	-	10	-	10	ns
	UB, LB High to Output High-Z	t _{BHZ}	-	10	-	10	-	10	ns

1. Refresh can not be implemented when t_{BSADV} is in the range of 0ns ~ 13ns. It may cause Refresh fail to use the device under that condition over 1.2μs without CS toggling. To avoid Refresh fail, 13ns for all frequency is needed for t_{BSADV}.

2. The High-Z timings measure a 100mV transition from either V_{OH} or V_{OL} toward V_{CCQ} x 0.5

3. The Low-Z timings measure a 100mV transition away from the High-Z level toward either V_{OH} or V_{OL}.

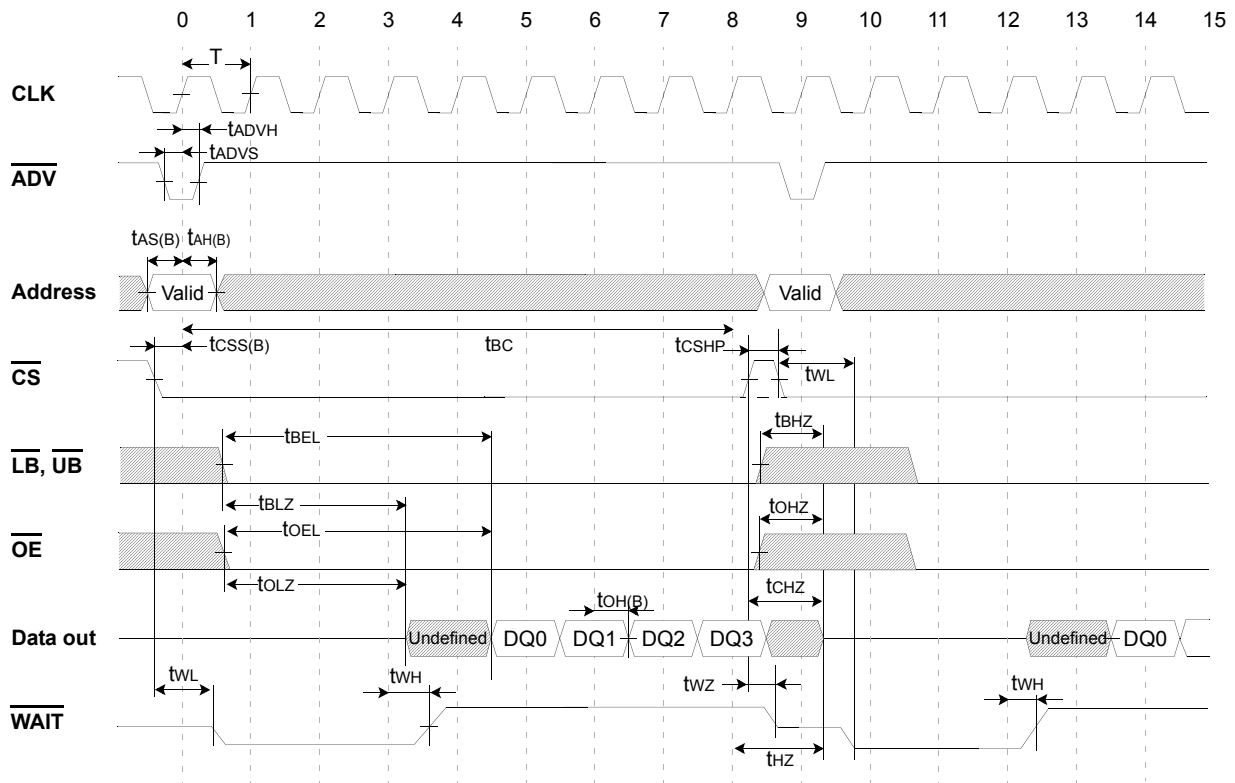
Parameter List		Symbol	Speed		Units
			Min	Max	
Asynch. Write	Write Cycle Time	t _{WC}	70	-	
	Chip Select to End of Write	t _{CW}	60	-	ns
	ADV Minimum Low Pulse Width	t _{ADV}	5	-	ns
	Address Set-up Time to Beginning of Write	t _{AS}	0	-	ns
	Address Set-up Time to ADV Rising	t _{AS(A)}	5	-	ns
	Address Hold Time from ADV Rising	t _{AH(A)}	3	-	ns
	CS Setup Time to ADV Rising	t _{CSS(A)}	5	-	ns
	Address Valid to End of Write	t _{AW}	60	-	ns
	UB, LB Valid to End of Write	t _{BW}	60	-	ns
	Write Pulse Width	t _{WP}	55 ¹⁾	-	ns
	Write Recovery Time	t _{WR}	0	-	ns
	Data to Write Time Overlap	t _{DW}	20	-	ns
	Data Hold from Write Time	t _{DH}	0	-	ns

1. t_{WP}(min)=70ns for continuous write longer than 1.2μs without CS toggling.

TIMING WAVEFORMS (SYNCH. READ / ASYNCH. WRITE)

Burst READ - Fixed Latency

(PS=VIH, WE=VIH, WAIT=High-Z, Latency=4, Burst Length=4, WP=Low enable, WC=one clock prior to the data)



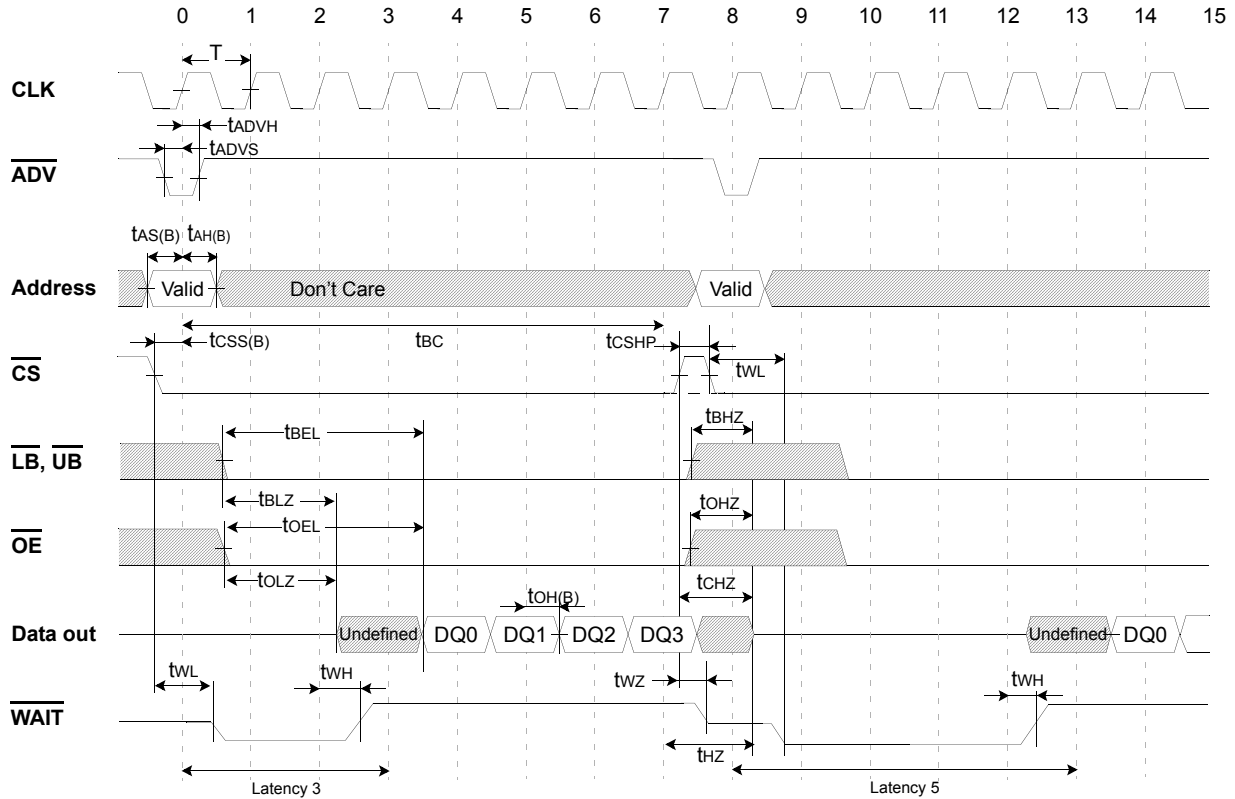
1. /WAIT Low(tWL) : Data not available(driven by $\overline{CS}$ low going edge or $\overline{ADV}$ low going edge)
 /WAIT High(tWH) : Data available(driven by Latency-1 clock)
 /WAIT High-Z(tWZ) : Data don't care(driven by $\overline{CS}$ high going edge)
2. Multiple clock risings are allowed during low ADV period. The data starts after set Latency from the last clock rising.
3. Burst operation should not be longer than tBC(1.2μs)

AC CHARACTERISTICS

Symbol	66MHz		80MHz		104MHz		Units	Symbol	66MHz		80MHz		104MHz		Units
	Min	Max	Min	Max	Min	Max			Min	Max	Min	Max	Min	Max	
T	15	200	12.5	200	9.6	200	ns	tBLZ	5	-	5	-	5	-	ns
tBC	-	1200	-	1200	-	1200	ns	tOLZ	5	-	5	-	5	-	ns
tADVS	3	-	3	-	3	-	ns	tHZ	-	10	-	10	-	10	ns
tADVH	2	-	2	-	2	-	ns	tCHZ	-	10	-	10	-	10	ns
tAS(B)	3	-	3	-	3	-	ns	tOHZ	-	10	-	10	-	10	ns
tAH(B)	2	-	2	-	2	-	ns	tBHZ	-	10	-	10	-	10	ns
tCSS(B)	3	-	3	-	3	-	ns	tCD	-	11	-	9	-	7	ns
tCSHP	5	-	5	-	5	-	ns	tOH(B)	2	-	2	-	2	-	ns
tBEL	20	-	20	-	20	-	ns	tWL	-	12	-	12	-	12	ns
tOEL	20	-	20	-	20	-	ns	tWH	-	11	-	9	-	7	ns
tWZ	-	10	-	10	-	10	ns								

Burst READ - Variable Latency

(PS=VIH, WE=VIH, WAIT=High-Z, Latency=3, Burst Length=4, WP=Low enable, WC=one clock prior to the data)



1. Delayed Latency should be taken increased when refresh is required. Refer to Latency Table.
2. /WAIT Low(tWL) : Data not available(driven by CS low going edge or ADV low going edge)
/WAIT High(tWH) : Data available(driven by Latency-1 clock)
/WAIT High-Z(tWZ) : Data don't care(driven by CS high going edge)
3. Multiple clock risings are allowed during low ADV period. The data starts after set Latency from the last clock rising.
4. Burst operation should not be longer than tBC(1.2μs)

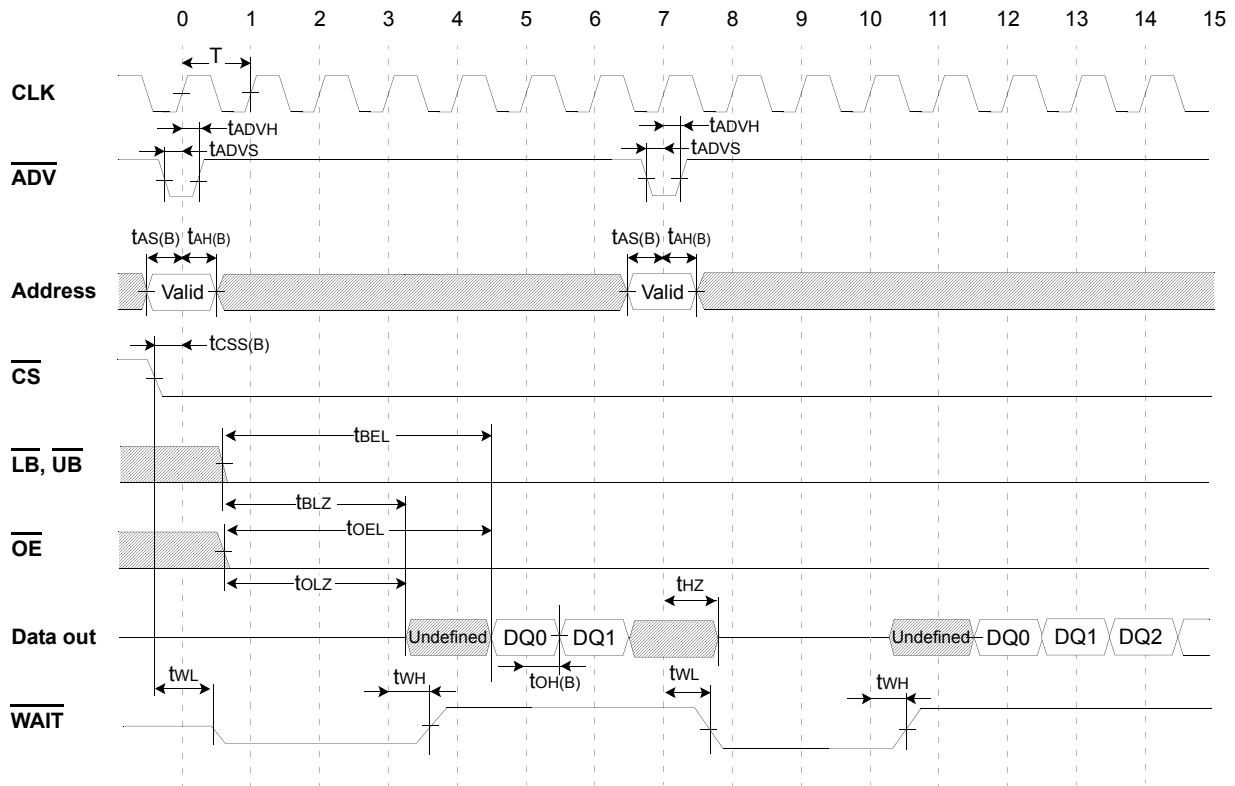
AC CHARACTERISTICS

Symbol	66MHz		80MHz		104MHz		Units	Symbol	66MHz		80MHz		104MHz		Units
	Min	Max	Min	Max	Min	Max			Min	Max	Min	Max	Min	Max	
T	15	200	12.5	200	9.6	200	ns	tBLZ	5	-	5	-	5	-	ns
tBC	-	1200	-	1200	-	1200	ns	tOLZ	5	-	5	-	5	-	ns
tADSV	3	-	3	-	3	-	ns	tHZ	-	10	-	10	-	10	ns
tADVH	2	-	2	-	2	-	ns	tCHZ	-	10	-	10	-	10	ns
tAS(B)	3	-	3	-	3	-	ns	tOHZ	-	10	-	10	-	10	ns
tAH(B)	2	-	2	-	2	-	ns	tBHZ	-	10	-	10	-	10	ns
tCSS(B)	3	-	3	-	3	-	ns	tCD	-	11	-	9	-	7	ns
tCSHP	5	-	5	-	5	-	ns	tOH(B)	2	-	2	-	2	-	ns
tBEL	20	-	20	-	20	-	ns	tWL	-	12	-	12	-	12	ns
tOEL	20	-	20	-	20	-	ns	tWH	-	11	-	9	-	7	ns
tWZ	-	10	-	10	-	10	ns								

K1B5616BAM

Burst READ (ADV Interrupt) - Fixed Latency

(PS=VIH, WE=VIH, WAIT=High-Z, Latency=4, Burst Length=4, WP=Low enable, WC=one clock prior to the data)



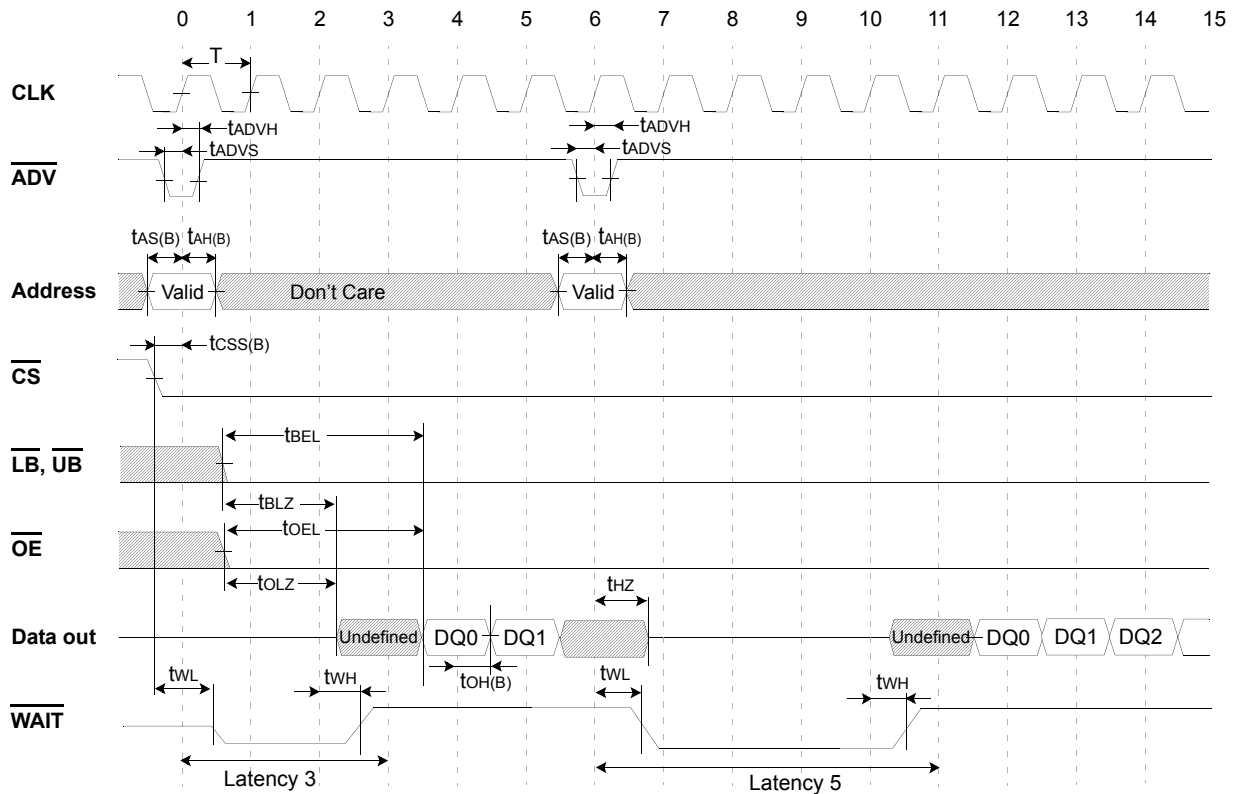
1. Refresh is blocked during $\overline{ADV}$ Interrupt Read and continuous Burst Read by ADV interrupt should not be longer than tBC (1.2us)
2. /WAIT Low(tWL) : Data not available(driven by $\overline{CS}$ low going edge or $\overline{ADV}$ low going edge)
/WAIT High(tWH) : Data available(driven by Latency-1 clock)
/WAIT High-Z(tWZ) : Data don't care(driven by $\overline{CS}$ high going edge)
3. Multiple clock risings are allowed during low $\overline{ADV}$ period but the First valid data come out after set Latency from the last clock rising.
4. Burst interrupt is allowable after the first data received by controller.

AC CHARACTERISTICS

Symbol	66MHz		80MHz		104MHz		Units	Symbol	66MHz		80MHz		104MHz		Units
	Min	Max	Min	Max	Min	Max			Min	Max	Min	Max	Min	Max	
T	15	200	12.5	200	9.6	200	ns	tBLZ	5	-	5	-	5	-	ns
tBC	-	1200	-	1200	-	1200	ns	tOLZ	5	-	5	-	5	-	ns
tADVS	3	-	3	-	3	-	ns	tHZ	-	10	-	10	-	10	ns
tADVH	2	-	2	-	2	-	ns	tCHZ	-	10	-	10	-	10	ns
tAS(B)	3	-	3	-	3	-	ns	tOHZ	-	10	-	10	-	10	ns
tAH(B)	2	-	2	-	2	-	ns	tBHZ	-	10	-	10	-	10	ns
tCSS(B)	3	-	3	-	3	-	ns	tCD	-	11	-	9	-	7	ns
tCSHP	5	-	5	-	5	-	ns	tOH(B)	2	-	2	-	2	-	ns
tBEL	20	-	20	-	20	-	ns	tWL	-	12	-	12	-	12	ns
tOEL	20	-	20	-	20	-	ns	tWH	-	11	-	9	-	7	ns
tWZ	-	10	-	10	-	10	ns								

Burst READ (ADV Interrupt) - Variable Latency

(PS=VIH, WE=VIH, WAIT=High-Z, Latency=3, Burst Length=4, WP=Low enable, WC=one clock prior to the data)



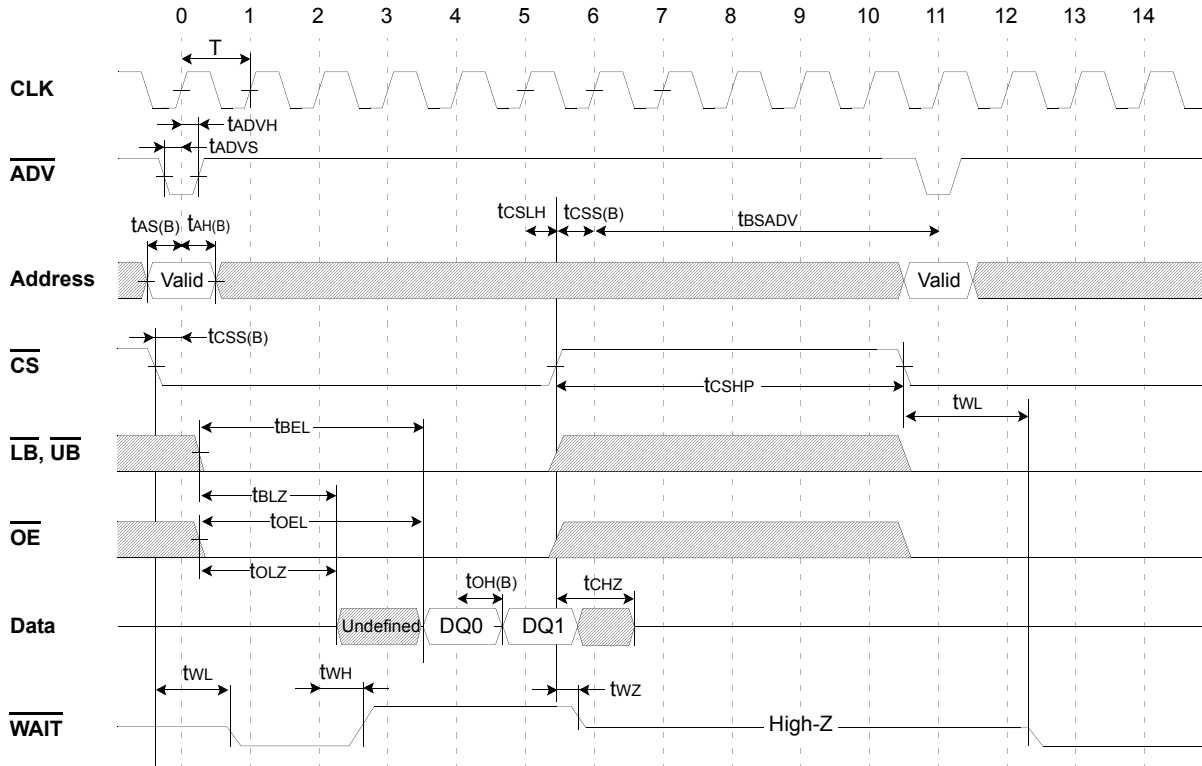
1. Delayed Latency should be taken increased when refresh is required. Refer to Latency Table.
2. Refresh is blocked during ADV Interrupt Read and continuous Burst Read by ADV interrupt should not be longer than tBC (1.2us)
3. /WAIT Low(tWL) : Data not available(driven by CS low going edge or ADV low going edge)
/WAIT High(tWH) : Data available(driven by Latency-1 clock)
/WAIT High-Z(tWZ) : Data don't care(driven by CS high going edge)
4. Multiple clock risings are allowed during low ADV period but the First valid data come out after set Latency from the last clock rising.
5. Burst interrupt is allowable after the first data received by controller.

AC CHARACTERISTICS

Symbol	66MHz		80MHz		104MHz		Units	Symbol	66MHz		80MHz		104MHz		Units
	Min	Max	Min	Max	Min	Max			Min	Max	Min	Max	Min	Max	
T	15	200	12.5	200	9.6	200	ns	tBLZ	5	-	5	-	5	-	ns
tBC	-	1200	-	1200	-	1200	ns	tOLZ	5	-	5	-	5	-	ns
tADVS	3	-	3	-	3	-	ns	tHZ	-	10	-	10	-	10	ns
tADVH	2	-	2	-	2	-	ns	tCHZ	-	10	-	10	-	10	ns
tAS(B)	3	-	3	-	3	-	ns	tOHZ	-	10	-	10	-	10	ns
tAH(B)	2	-	2	-	2	-	ns	tBHZ	-	10	-	10	-	10	ns
tCSS(B)	3	-	3	-	3	-	ns	tCD	-	11	-	9	-	7	ns
tCSHP	5	-	5	-	5	-	ns	tOH(B)	2	-	2	-	2	-	ns
tBEL	20	-	20	-	20	-	ns	tWL	-	12	-	12	-	12	ns
tOEL	20	-	20	-	20	-	ns	tWH	-	11	-	9	-	7	ns
tWZ	-	10	-	10	-	10	ns								

Burst READ STOP

(PS=VIH, Variable Latency=3, Burst Length=4, WP=Low Enable, WC=one clock prior to the data)

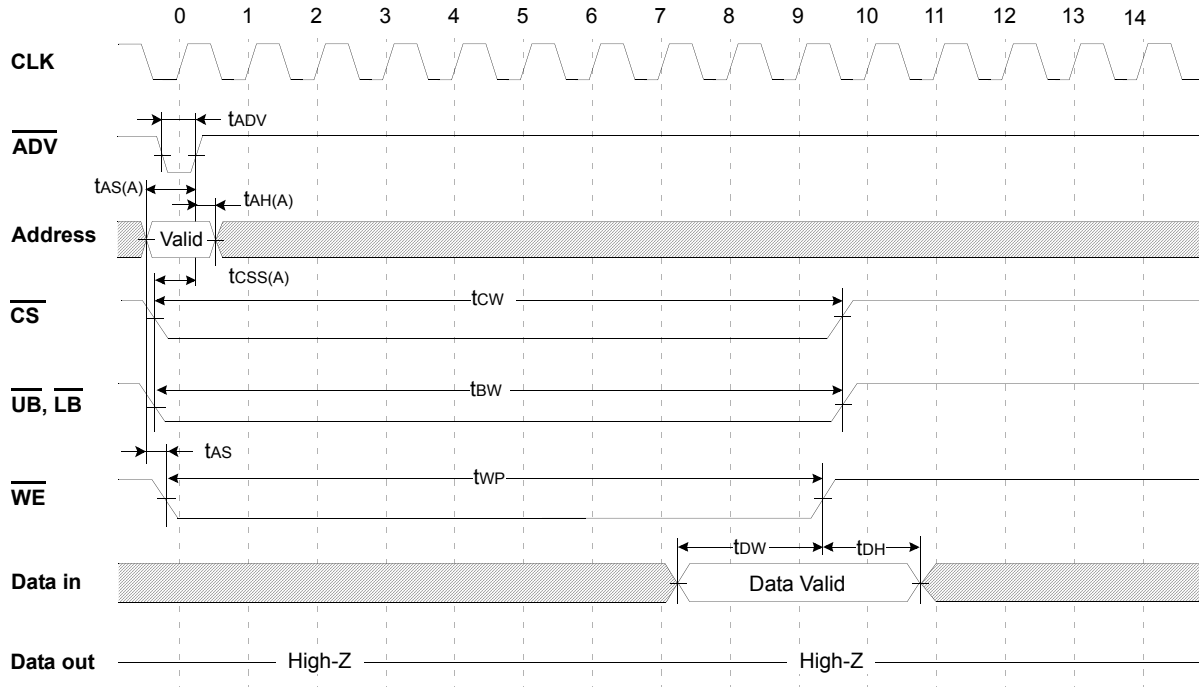


1. /WAIT Low(tWL) : Data not available(driven by $\overline{CS}$ low going edge or $\overline{ADV}$ low going edge)
/WAIT High(tWH) : Data available(driven by Latency-1 clock)
/WAIT High-Z(tWZ) : Data don't care(driven by $\overline{CS}$ high going edge)
2. Multiple clock risings are allowed during low ADV period. The data starts after set Latency from the last clock rising.
3. Refresh can not be implemented when tBSADV is in the range of 0ns ~ 13ns. It may cause Refresh fail to use the device under that condition over 1.2us without $\overline{CS}$ toggling. To avoid Refresh fail, 13ns for all frequency is needed for tBSADV.

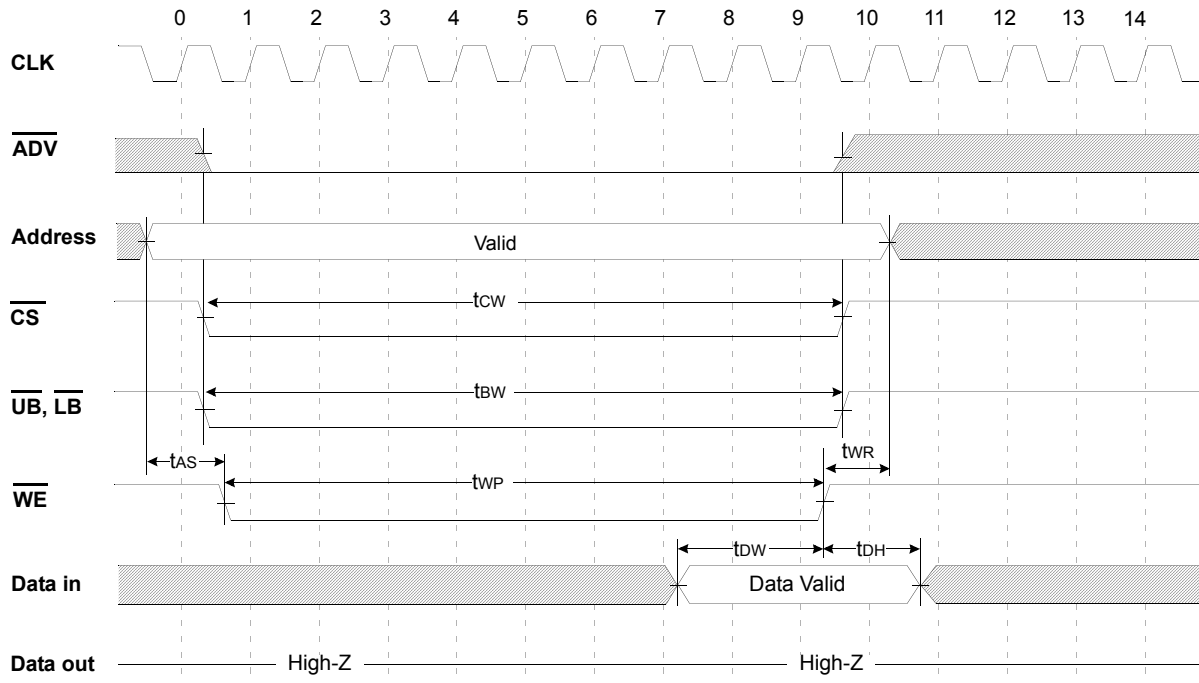
AC CHARACTERISTICS

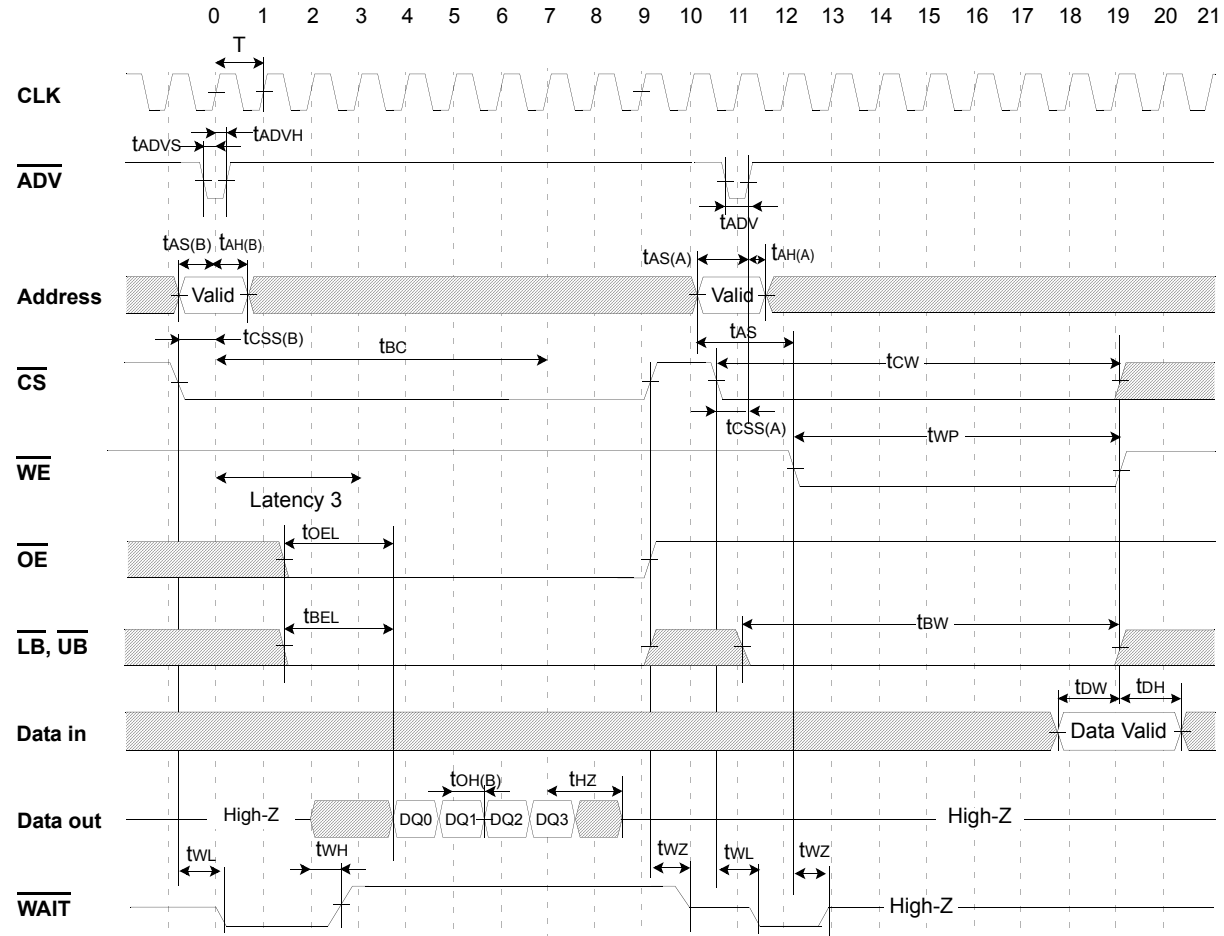
Symbol	66MHz		80MHz		104MHz		Units	Symbol	66MHz		80MHz		104MHz		Units
	Min	Max	Min	Max	Min	Max			Min	Max	Min	Max	Min	Max	
tBSADV	0	-	0	-	0	-	ns	tCD	-	11	-	9	-	7	ns
tCSLH	2	-	2	-	2	-	ns	tOH(B)	2	-	2	-	2	-	ns
tCSHP	5	-	5	-	5	-	ns	tCHZ	-	10	-	10	-	10	ns
tBEL	20	-	20	-	20	-	ns	tWL	-	12	-	12	-	12	ns
tOEL	20	-	20	-	20	-	ns	tWH	-	11	-	9	-	7	ns
tBLZ	5	-	5	-	5	-	ns	tWZ	-	10	-	10	-	10	ns
tOLZ	5	-	5	-	5	-	ns								

Asynch. WRITE (ADV Latch) (PS=VIH, OE=VIH, WAIT=High-Z)



Asynch. WRITE (ADV Fix Low) (PS=VIH, OE=VIH, WAIT=High-Z)

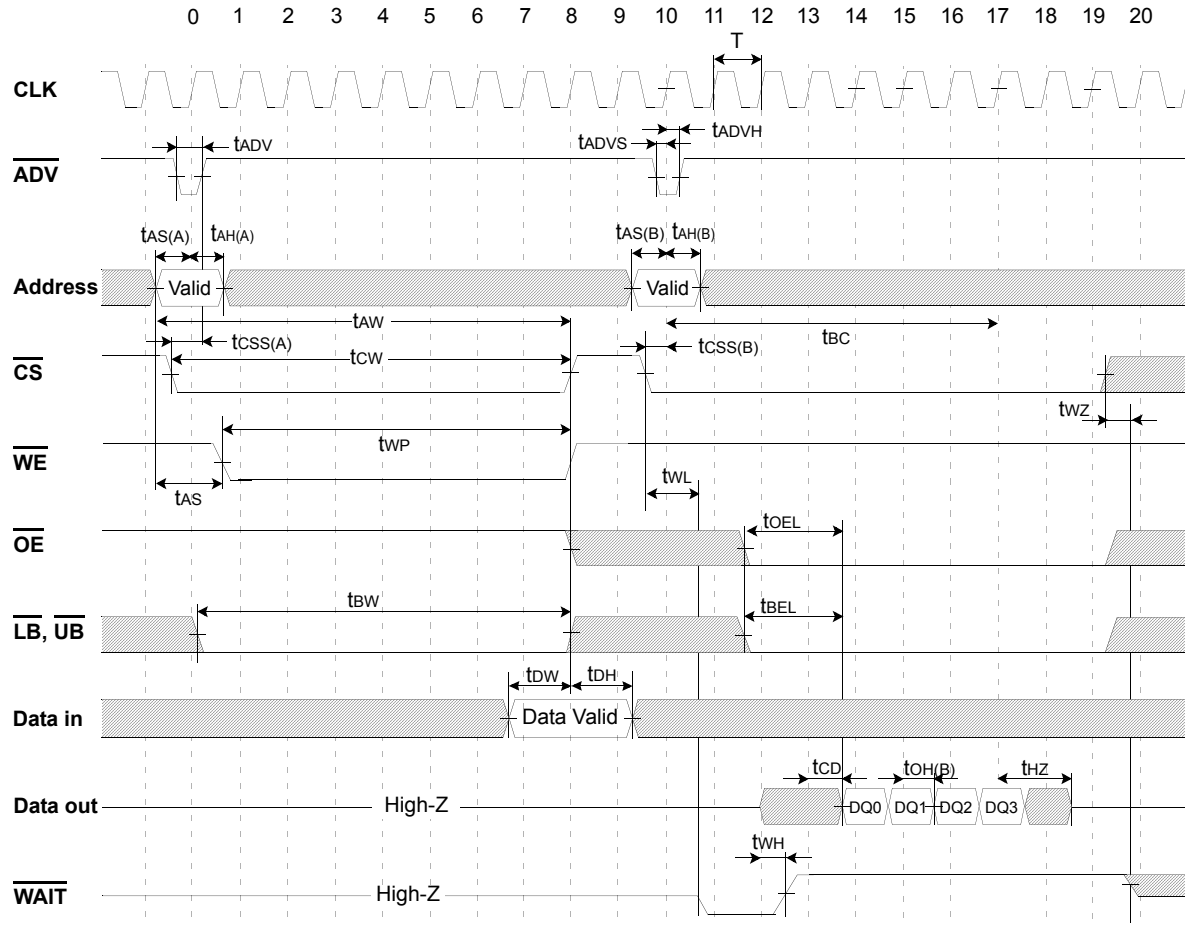


Burst READ followed by Asynch. WRITE
(PS=VIH, WAIT=High-Z, Variable Latency=3)

1. A write occurs during the overlap(t_{WP}) of low $\overline{CS}$ and low $\overline{WE}$. A write begins when $\overline{CS}$ goes low and $\overline{WE}$ goes low with asserting $\overline{UB}$ or $\overline{LB}$ for single byte operation or simultaneously asserting $\overline{UB}$ and $\overline{LB}$ for word operation. A write ends at the earliest transition when $\overline{CS}$ goes high or $\overline{WE}$ goes high. The t_{WP} is measured from the beginning of write to the end of write.
2. t_{BW} is measured from the address valid to the end of write. In this address latch type write timing, t_{WC} is same as t_{BW} .
3. t_{CW} is measured from the $\overline{CS}$ going low to the end of write.
4. t_{BW} is measured from the $\overline{UB}$ and $\overline{LB}$ going low to the end of write.

Asynch. WRITE followed by Burst READ

(PS=VIH, Variable Latency=3, Burst Length=4, WP=Low Enable, WC=one clock prior to the data)



1. /WAIT Low(t_{WL}) : Data not available(driven by $\overline{CS}$ low going edge or $\overline{ADV}$ low going edge)
 /WAIT High(t_{WH}) : Data available(driven by Latency-1 clock)
 /WAIT High-Z(t_{WZ}) : Data don't care(driven by $\overline{CS}$ high going edge)
2. Multiple clock risings are allowed during low $\overline{ADV}$ period. The data starts after set Latency from the last clock rising.
3. Burst operation should not be longer than t_{BC} (1.2 μ s)